MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Mansourah Massarah Gold Project
Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
3310 ROUGHER AND SCAVENGER FLOTATION
DESIGN OF MISCELLANEOUS FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RC	:	Reinforced Concrete

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1. INTENT

This document is to Prepare for design of miscellaneous foundations of **Rougher And Scavenger Flotation**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel & STAAD. For member strength calculation force values are directly taken from STAAD Output.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are directly taken from STAAD analysis of flotation structure and reference Document no MD-216-3310-EG-ST-CAL-2178. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

4.2 Load Combination

Load combinations are based on SBC-301,2007, MD-216-1000-EG-ST-CRT-1001, ACI 318M & ACI SP-17M(09)

Basic Load cases are –

1. Dead Load (D)
2. Fluid Load (FL1)
3. Live Load (L)
4. Seismic Load (SLX) (+/-)
5. Seismic Load (SLZ) (+/-)
6. Wind Load (WLX)(+/-)
7. Wind Load (WLZ)(+/-)

Allowable Stress Design

Load Case	Load Combination
201	D + F1
202	D + F1 + WLX
203	D + F1 + WLZ
204	D+F1+0.7SLX
205	D+F1+0.7SLZ
206	0.6 D + WLX
207	0.6 D + WLZ
208	0.6 (D+F1) + 0.7 SLX
209	0.6 (D+F1) + 0.7 SLZ
210	D+F1+L
211	D + F1 + O.75L + 0.75WLX
212	D+F1+0.75L+ 0.75WLZ
213	D+F1+0.75L+ 0.525SLX
214	D+F1+0.75L+ 0.525SLZ

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Ultimate Strength Design

Load Case	Load Combination
100	1.4 (D + F1)
101	1.2(D + F1) + 1.6 WLX
102	1.2(D + F1) + 1.6 WLZ
103	1.2(D + F1) + SLX
104	1.2(D + F1) + SLZ
105	0.9 D + 1.6 WLX
106	0.9 D + 1.6 WLZ
107	0.9 (D+ F1) + SLX
108	0.9 (D+ F1) + SLZ
109	1.2DL+1.2F1+1.6LL
110	1.2DL+1.6WLX+0.5LL
111	1.2DL+1.6WLZ+0.5LL
112	1.2DL+1.0SLX+0.5LL+0.5F1
113	1.2DL+1.0SLZ+0.5LL+0.5F1

4.3 Stability Against Overturning & Sliding

Overturning

Structures and building foundations, shall have the following minimum stability ratios against overturning:

- 1.5 for load combinations with wind loads
- 2 for other load combinations

Sliding

A minimum factor of safety against sliding shall be 1.5.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

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5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundation Type

Foundations shall be combined and isolated type. The design calculations as determined by the standard in Civil, Structural & Architectural design criteria (Doc no. MD-216-1000-EG-ST-CRT-1001). Refer Attachment 1 & 2 for combined foundation design.

5.4 Grouting

Grout shall be 30 mm thick cementitious under steel column bases of strength 40 Mpa.

5.5 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation Top & Column	50

5.6 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

5.7 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m^2 at 1.5m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5. DESIGN CALCULATION FOR FOUNDATION

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DESIGN OF FOUNDATION MKD. F1

Size of Columns	Along-X	Along-Z	ex	ez
Column A	0.40m	0.50m	-0.3000m	-2.90m
Column B	0.40m	0.50m	-0.3000m	2.90m
Size of fdn. raft:	1.50m	6.345m		
Thickness of raft:	0.40 m			
Original Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	0.35 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m^3			
Unit weight of soil:	18 kN/m^3			

Calculation of Weight of Foundation :

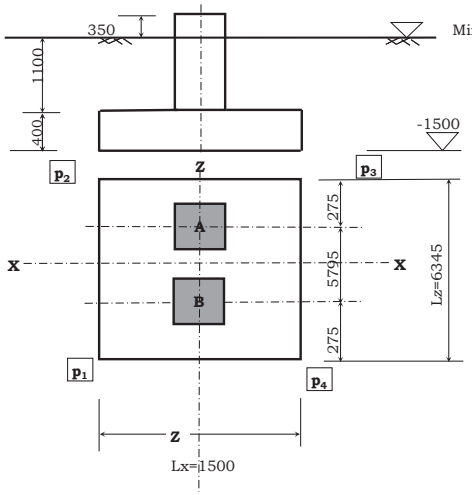
Weight of raft = (1.5*6.345*0.4*25) = 95.18KN

Weight of soil = ((1.5*6.345)-2*(0.4*0.5))*1.1*18 = 180.53KN

Total = 275.70KN

Design Input:

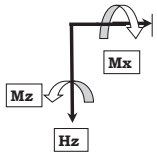
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%



Effective depth of slab along X direction = 336mm = 0.336m

Effective depth of slab along Z direction = 342mm = 0.342m

Clear projection along X direction = 0.8m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Basic Load Cases :

		Column A					Column B				
Sl. No	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kN-m)	Mx (kN-m)	V (kN)	Hx (kN)	Hz (kN)	Mz (kN-m)	Mx (kN-m)
1	Dead Load (D)	13	0.1	2.3	0.0	0.0	20.6	0.4	-2.3	0.0	0.0
2	Operating Fluid Load (FL1)	2.1	0.1	0.0	0.0	0.0	0.2	0.1	0.0	0.0	0.0
3	Live Load (LL)	49.4	-0.1	9.4	0.0	0.0	54.0	0.9	-9.4	0.0	0.0
4	Seismic Load (SLX)(+/-)	0.1	0.0	0.0	0.0	0.0	0.01	0.0	0.0	0.0	0.0
5	Seismic Load (SLZ)(+/-)	3.6	0.7	0.0	0.0	0.0	-0.6	-0.5	0.0	0.0	0.0
6	Wind Load (WLX) (+/-)	-14.0	0.0	-7.9	0.0	0.0	14.2	0.0	-7.2	0.0	0.0
7	Wind Load (WLZ) (+/-)	-6.0	-2.8	0.0	0.0	0.0	0.4	-3.7	0.0	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
	Allowable Stress Design										
201	D + F1	35.9	0.7	-10.8	0.0	16.5	311.60	0.70	-9.48	-0.01	16.50
202	D + F1 + WLX	36.1	0.7	-10.8	-15.1	98.2	311.80	0.70	-9.54	-15.11	70.27
203	D + F1 + WLZ	30.3	-5.8	-9.1	0.0	35.1	306.00	-5.80	-19.82	-0.01	35.04
204	D + F1+ 0.7 SLX	36.0	0.7	-10.8	0.0	16.3	311.68	0.70	-9.50	-0.01	16.31
205	D + F1+ 0.7 SLZ	38.0	0.9	-11.4	0.0	8.0	0.00	0.85	-9.83	-0.01	0.00
206	0.6 D + WLX	20.4	0.3	-6.1	-15.1	94.9	185.78	0.30	-5.55	-15.10	66.99
207	0.6 D + WLZ	14.6	-6.2	-4.8	0.0	28.5	179.98	-6.20	-16.25	0.00	28.45
208	0.6(D+F1)+0.7 SLX	21.6	0.4	-6.5	0.0	9.7	187.04	0.42	-5.71	-0.01	9.71
209	0.6(D+F1)+0.7 SLZ	23.6	0.3	-7.1	0.0	1.4	189.06	0.30	-6.54	-0.01	1.38
210	D + F1 + L	139.3	1.5	-41.8	0.0	29.8	415.00	1.50	-39.02	-0.01	29.82
211	D + F1 + O.75L + 0.75WLX	113.6	1.3	-34.1	-11.3	87.8	389.30	1.30	-31.68	-11.34	66.82
212	D+F1+0.75L+ 0.75WLZ	109.3	-3.6	-32.8	0.0	40.4	384.95	-3.58	-39.39	-0.01	40.40
213	D+F1+0.75L+ 0.525SLX	90.2	1.3	-34.1	0.0	26.4	365.94	1.30	-31.65	-0.01	26.35
214	D+F1+0.75L+ 0.525SLZ	91.8	1.5	-34.5	0.0	20.1	367.46	1.45	-31.83	-0.01	20.10
100	1.4 (D + F1)	50.3	1.0	-15.1	0.0	23.1	436.24	0.98	-13.27	-0.02	23.09
101	1.2(D + F1) + 1.6 WLX	43.4	0.8	-13.0	-24.2	150.6	374.24	0.84	-11.47	-24.17	105.83
102	1.2(D + F1) + 1.6 WLZ	34.1	-9.6	-10.2	0.0	49.5	364.96	-9.56	-27.92	-0.01	49.46
103	1.2(D + F1) + SLX	43.2	0.8	-13.0	0.0	19.6	374.03	0.84	-11.40	-0.01	19.53
104	1.2(D + F1) + SLZ	46.1	1.0	-13.8	0.0	7.6	376.92	1.04	-11.90	-0.01	7.62
105	0.9 D + 1.6 WLX	30.6	0.5	-11.0	-24.2	168.5	278.69	0.45	-10.19	-24.16	123.76
106	0.9 D + 1.6 WLZ	21.3	-10.0	-8.2	0.0	67.4	269.41	-9.95	-26.65	0.00	67.40
107	0.9 (D+ F1) + SLX	32.4	0.6	-9.7	0.0	14.6	280.55	0.63	-8.56	-0.01	14.59
108	0.9 (D+ F1) + SLZ	35.3	0.8	-10.6	0.0	2.7	283.44	0.83	-9.06	-0.01	2.68
109	1.2DL+1.2F1+1.6LL	208.5	2.1	-62.6	0.0	41.1	539.36	2.12	-58.63	-0.01	41.12
110	1.2DL+1.6WLX+0.5LL	92.3	1.0	-27.7	-24.2	163.8	423.18	1.00	-25.85	-24.16	119.13
111	1.2DL+1.6WLZ+0.5LL	83.1	-9.4	-24.9	0.0	62.8	413.90	-9.40	-42.31	0.00	62.76
112	1.2DL+1.0SLX+0.5LL+0.5F1	93.3	1.1	-28.0	0.0	30.1	424.12	1.10	-25.95	-0.01	30.07
113	1.2DL+1.0SLZ+0.5LL+0.5F1	96.2	1.3	-28.9	0.0	18.2	427.01	1.30	-26.45	-0.01	18.16

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D + F1	311.60	0.70	-9.48	-0.01	16.50	24.66	59.93	178.06	
202	D + F1 + WLX	311.80	0.70	-9.54	-15.11	70.27	24.53	14.08	178.17	7.43
203	D + F1 + WLZ	306.00	-5.80	-19.82	-0.01	35.04	11.58	27.71	21.10	
204	D + F1+ 0.7 SLX	311.68	0.70	-9.50	-0.01	16.31	24.61	60.61	178.10	
206	0.6 D + WLX	185.78	0.30	-5.55	-15.10	66.99	25.09	8.80	247.71	4.43
207	0.6 D + WLZ	179.98	-6.20	-16.25	0.00	28.45	8.31	20.07	11.61	
208	0.6(D+F1)+0.7 SLX	187.04	0.42	-5.71	-0.01	9.71	24.58	61.08	178.13	
209	0.6(D+F1)+0.7 SLZ	189.06	0.30	-6.54	-0.01	1.38	21.69	435.08	252.08	
210	D + F1 + L	415.00	1.50	-39.02	-0.01	29.82	7.98	44.15	110.67	
211	D + F1 + O.75L + 0.75WLX	389.30	1.30	-31.68	-11.34	66.82	9.22	18.48	119.78	12.36
212	D+F1+0.75L+ 0.75WLXZ	384.95	-3.58	-39.39	-0.01	40.40	7.33	30.23	43.07	
213	D+F1+0.75L+ 0.75WLXZ	365.94	1.30	-31.65	-0.01	26.35	8.67	44.05	112.60	
214	D+F1+0.75L+ 0.75WLXZ	367.46	1.45	-31.83	-0.01	20.10	8.66	57.99	101.37	

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DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

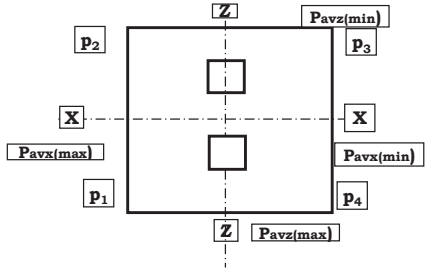
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 20 kN/m^2

Net safe bearing capacity of soil for DL, LL= 500 kN/m^2 For Wind load/Seis. load case 625 kN/m^2

Gross safe bearing capacity of soil for DL, LL= 527 kN/m^2 For Wind load/Seis. load case 652 kN/m^2



Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	D + F1	311.60	-9.48	16.50	32.74	3.98	1.64	38.36	35.08	30.40	27.12	11.36	36.72	28.76	34.38	31.10	**O.K**
202	D + F1 + WLX	311.80	-9.54	70.27	32.76	4.01	6.98	43.75	29.79	35.74	21.77	16.75	36.77	28.76	39.75	25.78	**O.K**
203	D + F1 + WLZ	306.00	-19.82	35.04	32.15	8.33	3.48	43.96	37.00	27.30	20.34	16.96	40.48	23.82	35.63	28.67	**O.K**
204	D + F1+ 0.7 SLX	311.68	-9.50	16.31	32.75	3.99	1.62	38.36	35.12	30.38	27.14	11.36	36.74	28.76	34.37	31.13	**O.K**
206	0.6 D + WLX	185.78	-5.55	66.99	19.52	2.33	6.66	28.51	15.20	23.84	10.53	1.51	21.86	17.19	26.18	12.87	**O.K**
207	0.6 D + WLZ	179.98	-16.25	28.45	18.91	6.83	2.83	28.57	22.91	14.91	9.25	1.57	25.74	12.08	21.74	16.08	**O.K**
208	0.6(D+F1)+0.7 SLX	187.04	-5.71	9.71	19.65	2.40	0.97	23.02	21.09	18.22	16.29	-3.98	22.06	17.26	20.62	18.69	**O.K**
209	0.6(D+F1)+0.7 SLZ	189.06	-6.54	1.38	19.86	2.75	0.14	22.75	22.47	17.25	16.98	-4.25	22.61	17.12	20.00	19.73	**O.K**
210	D + F1 + L	415.00	-39.02	29.82	43.60	16.40	2.96	62.96	57.04	30.17	24.24	35.96	60.00	27.21	46.57	40.64	**O.K**
211	D + F1 + O.75L + 0.75WLX	389.30	-31.68	66.82	40.90	13.31	6.64	60.86	47.58	34.23	20.95	33.86	54.22	27.59	47.55	34.27	**O.K**
212	D+F1+0.75L+ 0.75WLZ	384.95	-39.39	40.40	40.45	16.55	4.01	61.01	52.99	27.91	19.88	34.01	57.00	23.90	44.46	36.44	**O.K**
213	D+F1+0.75L+ 0.525SLX	365.94	-31.65	26.35	38.45	13.30	2.62	54.37	49.13	27.77	22.53	27.37	51.75	25.15	41.07	35.83	**O.K**
214	D+F1+0.75L+ 0.525SLZ	367.46	-31.83	20.10	38.61	13.38	2.00	53.98	49.99	27.23	23.24	26.98	51.99	25.24	40.61	36.62	**O.K**

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100	1.4 (D + F1)	436.24	-13.27	23.09	45.84	5.57	2.29	53.71	49.12	42.56	37.97	26.71	51.42	40.27	48.14	43.55	
101	1.2(D + F1) + 1.6 WLX	374.24	-11.47	105.83	39.32	4.82	10.52	54.66	33.62	45.02	23.99	27.66	44.14	34.51	49.84	28.81	
102	1.2(D + F1) + 1.6 WLZ	364.96	-27.92	49.46	38.35	11.74	4.91	55.00	45.17	31.53	21.70	28.00	50.09	26.62	43.27	33.44	
103	1.2(D + F1) + SLX	374.03	-11.40	19.53	39.30	4.79	1.94	46.03	42.15	36.45	32.57	19.03	44.09	34.51	41.24	37.36	
104	1.2(D + F1) + SLZ	376.92	-11.90	7.62	39.60	5.00	0.76	45.36	43.85	35.36	33.84	18.36	44.61	34.60	40.36	38.85	
105	0.9 D + 1.6 WLX	278.69	-10.19	123.76	29.28	4.28	12.30	45.86	21.27	37.30	12.70	18.86	33.57	25.00	41.58	16.99	
106	0.9 D + 1.6 WLZ	269.41	-26.65	67.40	28.31	11.20	6.70	46.20	32.81	23.80	10.41	19.20	39.51	17.11	35.00	21.61	
107	0.9 (D+ F1) + SLX	280.55	-8.56	14.59	29.48	3.60	1.45	34.52	31.63	27.33	24.43	7.52	33.08	25.88	30.93	28.03	
108	0.9 (D+ F1) + SLZ	283.44	-9.06	2.68	29.78	3.81	0.27	33.85	33.32	26.24	25.71	6.85	33.59	25.98	30.05	29.52	
109	1.2DL+1.2F1+1.6LL	539.36	-58.63	41.12	56.67	24.64	4.09	85.40	77.23	36.11	27.94	58.40	81.32	32.03	60.76	52.59	
110	1.2DL+1.6WLX+0.5LL	423.18	-25.85	119.13	44.46	10.87	11.84	67.16	43.49	45.43	21.76	40.16	55.33	33.60	56.30	32.63	
111	1.2DL+1.6WLZ+0.5LL	413.90	-42.31	62.76	43.49	17.78	6.24	67.51	55.03	31.94	19.47	40.51	61.27	25.71	49.73	37.25	
112	1.2DL+1.0SLX+0.5LL+0.5F1	424.12	-25.95	30.07	44.56	10.91	2.99	58.46	52.48	36.64	30.67	31.46	55.47	33.66	47.55	41.58	
113	1.2DL+1.0SLZ+0.5LL+0.5F1	427.01	-26.45	18.16	44.87	11.11	1.80	57.78	54.18	35.56	31.95	30.78	55.98	33.76	46.67	43.07	

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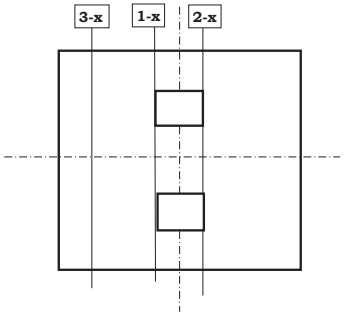
DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.4+2*0.5*0.342)+(0.5+2*0.5*0.342))*2
= 3.168 m
Punching Area= ((0.4+2*0.5*0.342)*(0.5+2*0.5*0.342))
= 0.62 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*25+1.1*18)
= 29.80 kN/m^2
Area of footing, A_f= 9.52 m^2



Overhang = 0.8 m
Effective depth = 0.336 m

Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
100	1.4 (D + F1)	45.47	46.22	47.97	6.28	3.98	9.23	142.68
101	1.2(D + F1) + 1.6 WLX	39.00	39.65	41.16	4.04	2.06	5.96	84.72
102	1.2(D + F1) + 1.6 WLZ	37.57	39.14	42.83	5.16	0.32	7.73	76.04
103	1.2(D + F1) + SLX	38.98	39.62	41.13	4.03	2.05	5.94	84.52
104	1.2(D + F1) + SLZ	39.27	39.94	41.51	4.17	2.11	6.15	87.22
105	0.9 D + 1.6 WLX	29.00	29.57	30.92	0.72	-1.05	1.13	-4.61
106	0.9 D + 1.6 WLZ	27.56	29.06	32.58	1.83	-2.79	2.90	-13.29
107	0.9 (D+ F1) + SLX	29.24	29.72	30.85	0.64	-0.84	1.01	-2.87
108	0.9 (D+ F1) + SLZ	29.53	30.04	31.24	0.78	-0.79	1.21	-0.17
109	1.2DL+1.2F1+1.6LL	55.03	58.32	66.07	13.68	3.52	20.37	239.08
110	1.2DL+1.6WLX+0.5LL	43.74	45.19	48.61	6.93	2.45	10.29	130.47
111	1.2DL+1.6WLZ+0.5LL	42.30	44.68	50.27	8.05	0.71	12.05	121.79
112	1.2DL+1.0SLX+0.5LL+0.5F1	43.84	45.29	48.72	6.97	2.48	10.35	131.35
113	1.2DL+1.0SLZ+0.5LL+0.5F1	44.13	45.61	49.11	7.11	2.53	10.55	134.05

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	13.68	0.12	0.16	0.000	129.63	604.80	12ϕT	187.00	300	150	753.98
At top	-2.79	0.02	0.03	0.000	0.00	302.40	12ϕT	374.00	300	250	452.39

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DESIGN OF FOUNDATION MKD. F1

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 20.37 kN/m

Shear stress , τ_v= (20.37*1000)/(1000*336)

= 0.06 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 239.08 kN

Strength of Concrete in shear Vc1 = 3673.46 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2162.73 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2099.12 kN **Table 22.6.5.2_ (c)**

Design Vc= 2099.12 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (239.08*1000)/(3168*342)

= 0.22 N/mm^2 **No shear reinforcement reqd.**

Calculation of distribution reinf in Z direction : Effective depth = 0.342 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	19.00	0.16	0.22	0.001	177.01	615.60	12ϕT	183.72	300	175	646.27
At top	167.33	1.43	1.91	0.005	1606.54	307.80	16ϕT	125.15	300	125	1608.50

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 115.33 kN/m

Shear stress , τ_v= (115.33*1000)/(1000*342)

= 0.34 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**12ϕT**	**150**	c/c
At Top	**12ϕT**	**250**	c/c
Along Z Direction :			
At Bottom	**12ϕT**	**175**	c/c
At Top	**16ϕT**	**125**	c/c

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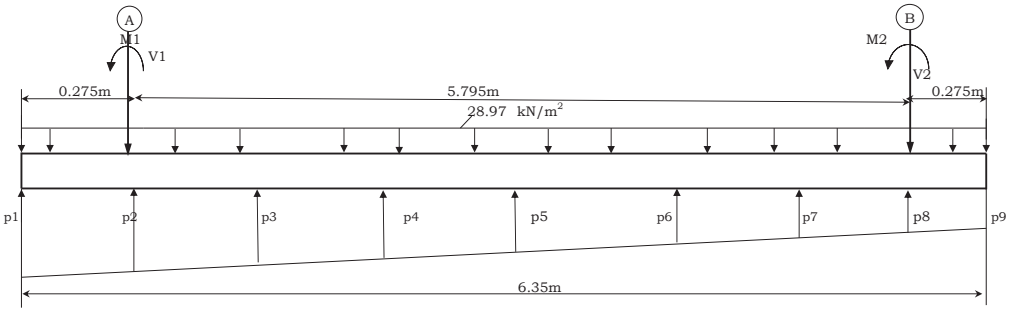
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DESIGN OF FOUNDATION MKD. F1

CALCULATION OF BM & SF :

(Along Z direction)



Load Case	Load Combination	Soil pressure in UDL (kN/m)								
		p1	p2	p3	p4	p5	p6	p7	p8	p9
100	1.4 (D + F1)	77.12								60.40
101	1.2(D + F1) + 1.6 WLX	66.21								51.76
102	1.2(D + F1) + 1.6 WLZ	75.13								39.92
105	0.9 D + 1.6 WLX	50.35								37.50
106	0.9 D + 1.6 WLZ	59.26								25.66
107	0.9 (D+ F1) + SLX	49.61			By STAAD					38.82
108	0.9 (D+ F1) + SLZ	50.38								38.96
109	1.2DL+1.2F1+1.6LL	121.97								48.04
110	1.2DL+1.6WLX+0.5LL	82.99								50.39
111	1.2DL+1.6WLZ+0.5LL	91.91								38.56
112	1.2DL+1.0SLX+0.5LL+0.5	83.21								50.48
113	1.2DL+1.0SLZ+0.5LL+0.5	83.97								50.63

From STAAD Output	
BM at top	251.00 (kN
BM at bot.	28.50 (kNm/m
SF (T. at top)	127.00 (kN/m)

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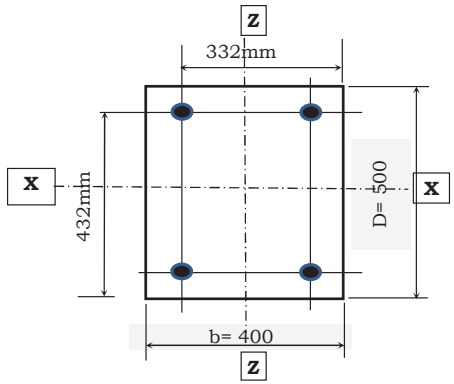
Load Case	Load Combination	Vertical Load (kN/m)							Moment (kNm/m)						
		V1	V2	V3	V4	V5	V6	V7	M1	M2	M3	M4	M5	M6	M7
100	1.4 (D + F1)	21.14			29.12				-13.27						
101	1.2(D + F1) + 1.6 WLX	-4.28			47.68				-11.47						
102	1.2(D + F1) + 1.6 WLZ	8.52			25.60				-27.92						
105	0.9 D + 1.6 WLX	-10.70			41.26				-10.19						
106	0.9 D + 1.6 WLZ	2.10			19.18				-26.65						
107	0.9 (D+ F1) + SLX	13.69			18.73				-8.56						
108	0.9 (D+ F1) + SLZ	17.19			18.12				-9.06						
109	1.2DL+1.2F1+1.6LL	97.16			111.360				-58.63						
110	1.2DL+1.6WLX+0.5LL	17.90			74.44				-25.85						
111	1.2DL+1.6WLZ+0.5LL	30.70			52.36				-42.31						
112	1.2DL+1.0SLX+0.5LL+0.5	41.45			51.83				-25.95						
113	1.2DL+1.0SLZ+0.5LL+0.5	44.95			51.22				-26.45						

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DESIGN OF COLUMN MKD **C1**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	16mm	
Dia. of link=	10mm	
Effective cover =	68mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1450mm
Unsupproted length of column about Z (l_{uz}) =	1450mm

For Columns not braced against sidesway,

Where, Effective length factor K= 1.2

Slenderness ratio,	$K\,l_{ux} / r_x$ =	12.06	< 22 SECTION 6.2.5a	Radius of gyration r_x=	144	mm
	$K\,l_{uz} / r_z$ =	15.07	< 22 SECTION 6.2.5a	Radius of gyration r_z=	115	mm

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Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	Dead Load (D)	13.0	0.1	2.3	0.0	0.0
2	Operating Fluid Load (FL1)	2.1	0.1	0.0	0.0	0.0
3	Live Load (LL)	49.4	-0.1	9.4	0.0	0.0
4	Seismic Load (SLX)(+/-)	0.1	0.0	0.0	0.0	0.0
5	Seismic Load (SLZ)(+/-)	3.6	0.7	0.0	0.0	0.0
6	Wind Load (WLX) (+/-)	-14.0	0.0	-7.9	0.0	0.0
7	Wind Load (WLZ) (+/-)	-6.0	-2.8	0.0	0.0	0.0

γ_y =	0.77
γ_z =	0.71

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
100	1.4 (D + F1)	2.11	0.02	0.5	3.3	0.0	0.7	0.005
101	1.2(D + F1) + 1.6 WLX	-0.4	0.02	-1.4	-0.7	0.0	2.2	-0.001
102	1.2(D + F1) + 1.6 WLZ	0.9	-0.6	0.4	1.3	1.0	0.6	0.002
103	1.2(D + F1) + SLX	1.8	0.02	0.4	2.8	0.0	0.6	0.004
104	1.2(D + F1) + SLZ	2.2	0.1	0.4	3.3	0.2	0.6	0.005
105	0.9 D + 1.6 WLX	-1.1	0.01	-1.5	-1.6	0.0	2.3	0.00
106	0.9 D + 1.6 WLZ	0.2	-0.7	0.3	0.3	1.0	0.5	0.00
107	0.9 (D+ F1) + SLX	1.4	0.01	0.3	2.1	0.0	0.5	0.00
108	0.9 (D+ F1) + SLZ	1.7	0.11	0.3	2.6	0.2	0.5	0.00
109	1.2DL+1.2F1+1.6LL	9.7	0.00	2.6	14.9	0.0	4.0	0.02
110	1.2DL+1.6WLX+0.5LL	1.8	0.01	-0.7	2.8	0.0	1.1	0.00
111	1.2DL+1.6WLZ+0.5LL	3.1	-0.7	1.1	4.7	1.0	1.7	0.01
112	1.2DL+1.0SLX+0.5LL+0.5F1	4.1	0.01	1.1	6.4	0.0	1.7	0.01
113	1.2DL+1.0SLZ+0.5LL+0.5F1	4.5	0.1	1.1	6.9	0.2	1.7	0.01

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Reinforcement: Provide,

4-Nos	16 mm Dia
4-Nos	16 mm Dia
3-Nos	16 mm Dia

804.57 Sq mm (At Corner)
804.57 Sq mm (Along Z axis)
603.43 Sq mm (Along X axis)

A_{sc} pvd. = 2212.57 Sq mm

Total A_{sc} = 1.11 %

β = P_n / P_0 (range is 0.55 to 0.70)

Where, $P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 681.3456 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
100	0.05	0.05	14.0	17.5	0.55	1.16	0.0256	O.K
101	0.04	0.04	11.2	14.0	0.55	1.16	0.12	O.K
102	0.05	0.05	14.0	17.5	0.55	1.16	0.07	O.K
103	0.05	0.05	14.0	17.5	0.55	1.16	0.02	O.K
104	0.05	0.05	14.0	17.5	0.55	1.16	0.03	O.K
105	0.05	0.05	14.0	17.5	0.55	1.16	0.10	O.K
106	0.05	0.05	14.0	17.5	0.55	1.16	0.06	O.K
107	0.05	0.05	14.0	17.5	0.55	1.16	0.02	O.K
108	0.05	0.05	14.0	17.5	0.55	1.16	0.02	O.K
109	0.05	0.05	14.0	17.5	0.55	1.16	0.18	O.K
110	0.05	0.05	14.0	17.5	0.55	1.16	0.04	O.K
111	0.05	0.05	14.0	17.5	0.55	1.16	0.11	O.K
112	0.05	0.05	14.0	17.5	0.55	1.16	0.07	O.K
113	0.05	0.05	14.0	17.5	0.55	1.16	0.07	O.K

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DESIGN OF FOUNDATION MKD. F2 (STAAD Node -9572)

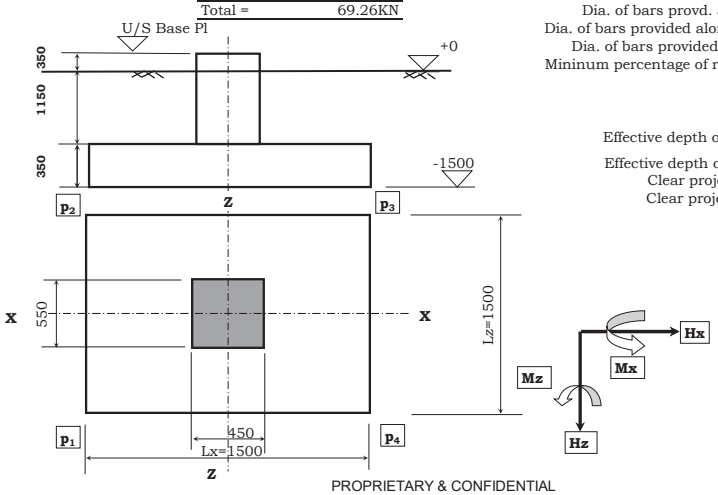
	Along-X	Along-Z
Size of column :	0.45m	0.55m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.35 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Fluid Load	FL1
Max. Fluid Load	FL2
Wind Load	WL
Seismic Load	SL

Calculation of Weight of Foundation :

Weight of raft & Col.=	(1.5*1.5*0.35*24)+(0.45*0.55*1.5*24) =	27.81KN
Weight of soil =	((1.5*1.5)-(0.45*0.55))*1.15*18 =	41.45KN
	Total =	69.26KN



Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 294mm = 0.294m

Effective depth of slab along Z direction = 282mm = 0.282m

Clear projection along Z direction = 0.475m

Clear projection along X direction = 0.525m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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DESIGN OF FOUNDATION MKD. F2

Combined Load Cases:		Load at top of foundation					Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201:1.0DL+1.0FL1		12.8	0.0	0.0	-0.8	0.0	82.01	0.00	0.00	-0.84	-1.55
202:1.0DL+1.0FL2		12.7	0.0	0.0	-0.8	0.0	81.91	0.00	0.00	-0.83	-1.54
203:1.0DL+1.0FL1+1.0LL+1.0TL		42.9	0.0	0.0	-3.5	0.0	112.17	0.00	0.00	-3.52	-6.51
204:1.0DL+1.0FL2+1.0LL+1.0TL		42.8	0.0	0.0	-3.5	0.0	112.07	0.00	0.00	-3.51	-6.49
205:1.0DL+1.0FL1+0.75LL+0.75TL		35.4	0.0	0.0	-2.9	0.0	104.63	0.00	0.00	-2.85	-5.27
206:1.0DL+1.0FL2+0.75LL+0.75TL		35.3	0.0	0.0	-2.8	0.0	104.53	0.00	0.00	-2.84	-5.25
207:1.0DL+1.0FL1+1.0PL		21.8	0.0	0.0	-3.2	0.0	91.09	0.00	0.00	-3.23	-5.98
208:1.0DL+1.0FL2+1.0PL		21.7	0.0	0.0	-3.2	0.0	90.99	0.00	0.00	-3.22	-5.96
209:1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL		52.0	0.0	0.0	-5.9	0.0	121.25	0.00	0.00	-5.90	-10.92
210:1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL		51.9	0.0	0.0	-5.9	0.0	121.15	0.00	0.00	-5.90	-10.92
211:1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL		44.5	0.0	0.0	-5.2	0.0	113.71	0.00	0.00	-5.24	-9.69
212:1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL		44.4	0.0	0.0	-5.2	0.0	113.61	0.00	0.00	-5.23	-9.68
213:1.0DL+1.0FL1+1.0WLX1		15.3	-0.9	0.0	-2.2	0.0	84.54	-0.93	-1.72	-2.18	-4.03
214:1.0DL+1.0FL1+1.0WLX2		10.2	0.9	0.0	0.5	0.0	79.48	0.93	1.72	0.49	0.91
215:1.0DL+1.0FL1+1.0WLZ1		21.2	0.0	0.0	-4.5	0.0	90.44	0.00	0.00	-4.52	-8.36
216:1.0DL+1.0FL1+1.0WLZ2		4.3	0.0	0.0	2.8	0.0	73.58	0.00	0.00	2.83	5.24
217:1.0DL+1.0FL2+1.0WLX1		15.2	-0.9	0.0	-2.2	0.0	84.44	-0.93	-1.72	-2.17	-4.01
218:1.0DL+1.0FL2+1.0WLX2		10.1	0.9	0.0	0.5	0.0	79.38	0.93	1.72	0.50	0.93
219:1.0DL+1.0FL2+1.0WLZ1		21.1	0.0	0.0	-4.5	0.0	90.34	0.00	0.00	-4.51	-8.34
220:1.0DL+1.0FL2+1.0WLZ2		4.2	0.0	0.0	2.8	0.0	73.48	0.00	0.00	2.84	5.25
221:1.0DL+1.0FL1+0.7SLX1		13.7	0.0	0.0	-1.3	0.0	82.92	0.00	0.00	-1.31	-2.42
222:1.0DL+1.0FL1+0.7SLX2		11.8	0.0	0.0	-0.4	0.0	81.09	0.01	0.02	-0.37	-0.68
223:1.0DL+1.0FL1+0.7SLZ1		15.5	0.0	0.0	-1.7	0.0	84.78	0.00	0.00	-1.73	-3.20
224:1.0DL+1.0FL1+0.7SLZ2		10.0	0.0	0.0	0.0	0.0	79.24	0.00	0.00	0.04	0.07
225:1.0DL+1.0FL2+0.7SLX1		13.6	0.0	0.0	-1.3	0.0	82.82	0.00	0.00	-1.30	-2.41
226:1.0DL+1.0FL2+0.7SLX2		11.7	0.0	0.0	-0.4	0.0	80.99	0.01	0.02	-0.37	-0.68
227:1.0DL+1.0FL2+0.7SLZ1		15.4	0.0	0.0	-1.7	0.0	84.68	0.00	0.00	-1.72	-3.18
228:1.0DL+1.0FL2+0.7SLZ2		9.9	0.0	0.0	0.1	0.0	79.14	0.00	0.00	0.05	0.09
229:1.0DL+1.0FL1+0.75WLX1+0.75LL		37.2	-0.7	0.0	-3.6	0.0	106.46	-0.69	-1.28	-3.59	-6.64
230:1.0DL+1.0FL1+0.75WLX2+0.75LL		33.4	0.7	0.0	-1.6	0.0	102.67	0.70	1.30	-1.59	-2.94
231:1.0DL+1.0FL1+0.75WLZ1+0.75LL		41.6	0.0	0.0	-5.4	0.0	110.88	0.00	0.00	-5.35	-9.90
232:1.0DL+1.0FL1+0.75WLZ2+0.75LL		29.0	0.0	0.0	0.2	0.0	98.24	0.00	0.00	0.16	0.30

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233:1.0DL+1.0FL2+0.75WLX1+0.75LL	37.1	-0.7	0.0	-3.6	0.0	106.36	-0.69	-1.28	-3.59	-6.64
234:1.0DL+1.0FL2+0.75WLX2+0.75LL	33.3	0.7	0.0	-1.6	0.0	102.57	0.70	1.30	-1.58	-2.92
235:1.0DL+1.0FL2+0.75WLZ1+0.75LL	41.5	0.0	0.0	-5.3	0.0	110.79	0.00	0.00	-5.34	-9.88
236:1.0DL+1.0FL2+0.75WLZ2+0.75LL	28.9	0.0	0.0	0.2	0.0	98.14	0.00	0.00	0.17	0.31
237:1.0DL+1.0FL1+0.525SLX1+0.75LL	36.0	0.0	0.0	-2.9	0.0	105.25	0.00	0.00	-2.94	-5.44
238:1.0DL+1.0FL1+0.525SLX2+0.75LL	34.6	0.0	0.0	-2.2	0.0	103.88	0.01	0.02	-2.24	-4.14
239:1.0DL+1.0FL1+0.525SLZ1+0.75LL	37.4	0.0	0.0	-3.3	0.0	106.64	0.00	0.00	-3.26	-6.03
240:1.0DL+1.0FL1+0.525SLZ2+0.75LL	33.2	0.0	0.0	-1.9	0.0	102.49	0.00	0.00	-1.93	-3.57
241:1.0DL+1.0FL2+0.525SLX1+0.75LL	35.9	0.0	0.0	-2.9	0.0	105.15	0.00	0.00	-2.93	-5.42
242:1.0DL+1.0FL2+0.525SLX2+0.75LL	34.5	0.0	0.0	-2.2	0.0	103.78	0.01	0.02	-2.23	-4.13
243:1.0DL+1.0FL2+0.525SLZ1+0.75LL	37.3	0.0	0.0	-3.3	0.0	106.54	0.00	0.00	-3.25	-6.01
244:1.0DL+1.0FL2+0.525SLZ2+0.75LL	33.1	0.0	0.0	-1.9	0.0	102.39	0.00	0.00	-1.92	-3.55
245:0.6DL+1.0WLX1	10.4	-0.9	0.0	-1.9	0.0	79.70	-0.93	-1.72	-1.87	-3.46
246:0.6DL+1.0WLX2	5.4	0.9	0.0	0.8	0.0	74.64	0.93	1.72	0.80	1.48
247:0.6DL+1.0WLZ1	16.3	0.0	0.0	-4.2	0.0	85.59	0.00	0.00	-4.21	-7.79
248:0.6DL+1.0WLZ2	-0.5	0.0	0.0	3.1	0.0	68.74	0.00	0.00	3.14	5.81
249:0.6DL+0.6FL1+0.7SLX1	8.6	0.0	0.0	-1.0	0.0	77.82	0.00	0.00	-0.98	-1.81
250:0.6DL+0.6FL1+0.7SLX2	6.7	0.0	0.0	0.0	0.0	75.99	0.01	0.02	-0.04	-0.07
251:0.6DL+0.6FL1+0.7SLZ1	10.4	0.0	0.0	-1.4	0.0	79.68	0.00	0.00	-1.39	-2.57
252:0.6DL+0.6FL1+0.7SLZ2	4.9	0.0	0.0	0.4	0.0	74.14	0.00	0.00	0.38	0.70
Ultimate Conditions										
101:1.4DL+1.4FL1	17.9	0.0	0.0	-1.2	0.0	87.11	0.00	0.00	-1.18	-2.18
102:1.4DL+1.4FL2	17.7	0.0	0.0	-1.2	0.0	86.97	0.00	0.00	-1.17	-2.16
103:1.2DL+1.2FL1+1.6LL	63.4	0.0	0.0	-4.7	0.0	132.68	0.00	0.00	-4.74	-8.77
104:1.2DL+1.2FL2+1.6LL	63.3	0.0	0.0	-4.7	0.0	132.56	0.00	0.00	-4.73	-8.75
105:1.2DL+1.2FL1+1.2TL+1.6LL	63.5	0.0	0.0	-5.2	0.0	132.78	0.00	0.00	-5.16	-9.55
106:1.2DL+1.2FL2+1.2TL+1.6LL	63.4	0.0	0.0	-5.2	0.0	132.66	0.00	0.00	-5.15	-9.53
107:1.4DL+1.4FL1+1.4PL	30.6	0.0	0.0	-4.5	0.0	99.82	0.00	0.00	-4.52	-8.36
108:1.4DL+1.4FL2+1.4PL	30.4	0.0	0.0	-4.5	0.0	99.68	0.00	0.00	-4.50	-8.33
109:1.2DL+1.2FL1+1.6LL+1.2PL	74.3	0.0	0.0	-7.6	0.0	143.57	0.00	0.00	-7.60	-14.06
110:1.2DL+1.2FL2+1.6LL+1.2PL	74.2	0.0	0.0	-7.6	0.0	143.45	0.00	0.00	-7.59	-14.04
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	74.4	0.0	0.0	-8.0	0.0	143.68	0.00	0.00	-8.02	-14.84
112:1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	74.3	0.0	0.0	-8.0	0.0	143.56	0.00	0.00	-8.01	-14.82
113:1.2DL+0.8WLX1	17.8	-0.7	0.0	-2.1	0.0	87.10	-0.74	-1.37	-2.13	-3.94
114:1.2DL+0.8WLX2	13.8	0.7	0.0	0.0	0.0	83.05	0.74	1.37	0.00	0.00
115:1.2DL+0.8WLZ1	22.6	0.0	0.0	-4.0	0.0	91.82	0.00	0.00	-4.00	-7.40
116:1.2DL+0.8WLZ2	9.1	0.0	0.0	1.9	0.0	78.33	0.00	0.00	1.87	3.46
117:1.2DL+1.6WLX1+0.5LL	34.9	-1.5	0.0	-4.4	0.0	104.16	-1.48	-2.74	-4.37	-8.08
118:1.2DL+1.6WLX2+0.5LL	26.8	1.5	0.0	-0.1	0.0	96.06	1.48	2.74	-0.09	-0.17
119:1.2DL+1.6WLZ1+0.5LL	44.3	0.0	0.0	-8.1	0.0	113.60	0.00	0.00	-8.11	-15.00
120:1.2DL+1.6WLZ2+0.5LL	17.4	0.0	0.0	3.7	0.0	86.63	0.00	0.00	3.65	6.75
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	31.9	0.0	0.0	-2.9	0.0	101.20	-0.01	-0.02	-2.88	-5.33
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	29.3	0.0	0.0	-1.5	0.0	98.59	0.01	0.02	-1.54	-2.85
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	34.6	0.0	0.0	-3.5	0.0	103.86	0.00	0.00	-3.48	-6.44
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	26.7	0.0	0.0	-0.9	0.0	95.94	0.00	0.00	-0.94	-1.74
125:0.9DL+1.6WLX1	15.9	-1.5	0.0	-2.9	0.0	85.17	-1.48	-2.74	-2.94	-5.44
126:0.9DL+1.6WLX2	7.8	1.5	0.0	1.3	0.0	77.07	1.48	2.74	1.34	2.48
127:0.9DL+1.6WLZ1	25.3	0.0	0.0	-6.7	0.0	94.60	0.00	0.00	-6.68	-12.36
128:0.9DL+1.6WLZ2	-1.6	0.0	0.0	5.1	0.0	67.64	0.00	0.00	5.08	9.40
129:0.9DL+1.0SLX1	13.2	0.0	0.0	-1.5	0.0	82.43	-0.01	-0.02	-1.47	-2.72
130:0.9DL+1.0SLX2	10.6	0.0	0.0	-0.1	0.0	79.81	0.01	0.02	-0.13	-0.24
131:0.9DL+1.0SLZ1	15.8	0.0	0.0	-2.1	0.0	85.08	0.00	0.00	-2.07	-3.83
132:0.9DL+1.0SLZ2	7.9	0.0	0.0	0.5	0.0	77.16	0.00	0.00	0.47	0.87

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Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0FL1	82.0	0.0	0.0	-0.8	-1.6		39.6		39.1
202	1.0DL+1.0FL2	81.9	0.0	0.0	-0.8	-1.5		40.0		39.5
203	1.0DL+1.0FL1+1.0LL+1.0TL	112.2	0.0	0.0	-3.5	-6.5		12.9		12.7
204	1.0DL+1.0FL2+1.0LL+1.0TL	112.1	0.0	0.0	-3.5	-6.5		12.9		12.8
205	1.0DL+1.0FL1+0.75LL+0.75TL	104.6	0.0	0.0	-2.9	-5.3		14.9		14.7
206	1.0DL+1.0FL2+0.75LL+0.75TL	104.5	0.0	0.0	-2.8	-5.3		14.9		14.7
207	1.0DL+1.0FL1+1.0PL	91.1	0.0	0.0	-3.2	-6.0		11.4		11.3
208	1.0DL+1.0FL2+1.0PL	91.0	0.0	0.0	-3.2	-6.0		11.5		11.3
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0PL	121.3	0.0	0.0	-5.9	-10.9		8.3		8.2
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0PL	121.2	0.0	0.0	-5.9	-10.9		8.3		8.2
211	1.0DL+1.0FL1+0.75LL+0.75TL+1.0PL	113.7	0.0	0.0	-5.2	-9.7		8.8		8.7
212	1.0DL+1.0FL2+0.75LL+0.75TL+1.0PL	113.6	0.0	0.0	-5.2	-9.7		8.8		8.7
213	1.0DL+1.0FL1+1.0WLX1	84.5	-0.9	-1.7	-2.2	-4.0	36.9	15.7	36.4	15.5
214	1.0DL+1.0FL1+1.0WLX2	79.5	0.9	1.7	0.5	0.9	34.6	65.8	34.2	64.9
215	1.0DL+1.0FL1+1.0WLZ1	90.4	0.0	0.0	-4.5	-8.4		8.1		8.0
216	1.0DL+1.0FL1+1.0WLZ2	73.6	0.0	0.0	2.8	5.2		10.5		10.4
217	1.0DL+1.0FL2+1.0WLX1	84.4	-0.9	-1.7	-2.2	-4.0	36.8	15.8	36.3	15.6
218	1.0DL+1.0FL2+1.0WLX2	79.4	0.9	1.7	0.5	0.9	34.6	64.4	34.1	63.5
219	1.0DL+1.0FL2+1.0WLZ1	90.3	0.0	0.0	-4.5	-8.3		8.1		8.0
220	1.0DL+1.0FL2+1.0WLZ2	73.5	0.0	0.0	2.8	5.3		10.5		10.3
221	1.0DL+1.0FL1+0.7SLX1	82.9	0.0	0.0	-1.3	-2.4		25.7		25.3
222	1.0DL+1.0FL1+0.7SLX2	81.1	0.0	0.0	-0.4	-0.7		88.8		87.7
223	1.0DL+1.0FL1+0.7SLZ1	84.8	0.0	0.0	-1.7	-3.2		19.9		19.6
224	1.0DL+1.0FL1+0.7SLZ2	79.2	0.0	0.0	0.0	0.1		803.1		792.4
225	1.0DL+1.0FL2+0.7SLX1	82.8	0.0	0.0	-1.3	-2.4		25.8		25.5
226	1.0DL+1.0FL2+0.7SLX2	81.0	0.0	0.0	-0.4	-0.7		88.7		87.6
227	1.0DL+1.0FL2+0.7SLZ1	84.7	0.0	0.0	-1.7	-3.2		20.0		19.7
228	1.0DL+1.0FL2+0.7SLZ2	79.1	0.0	0.0	0.1	0.1		641.7		633.1
229	1.0DL+1.0FL1+0.75WLX1+0.75LL	106.5	-0.7	-1.3	-3.6	-6.6	62.5	12.0	61.7	11.9
230	1.0DL+1.0FL1+0.75WLX2+0.75LL	102.7	0.7	1.3	-1.6	-2.9	59.5	26.2	58.7	25.8
231	1.0DL+1.0FL1+0.75WLZ1+0.75LL	110.9	0.0	0.0	-5.4	-9.9		8.4		8.3
232	1.0DL+1.0FL1+0.75WLZ2+0.75LL	98.2	0.0	0.0	0.2	0.3		248.9		245.6
233	1.0DL+1.0FL2+0.75WLX1+0.75LL	106.4	-0.7	-1.3	-3.6	-6.6	62.5	12.0	61.7	11.9
234	1.0DL+1.0FL2+0.75WLX2+0.75LL	102.6	0.7	1.3	-1.6	-2.9	59.4	26.3	58.6	26.0
235	1.0DL+1.0FL2+0.75WLZ1+0.75LL	110.8	0.0	0.0	-5.3	-9.9		8.4		8.3
236	1.0DL+1.0FL2+0.75WLZ2+0.75LL	98.1	0.0	0.0	0.2	0.3		234.0		230.9
237	1.0DL+1.0FL1+0.525SLX1+0.75LL	105.3	0.0	0.0	-2.9	-5.4		14.5		14.3
238	1.0DL+1.0FL1+0.525SLX2+0.75LL	103.9	0.0	0.0	-2.2	-4.1		18.8		18.6
239	1.0DL+1.0FL1+0.525SLZ1+0.75LL	106.6	0.0	0.0	-3.3	-6.0		13.3		13.1
240	1.0DL+1.0FL1+0.525SLZ2+0.75LL	102.5	0.0	0.0	-1.9	-3.6		21.5		21.2
241	1.0DL+1.0FL2+0.525SLX1+0.75LL	105.2	0.0	0.0	-2.9	-5.4		14.5		14.4
242	1.0DL+1.0FL2+0.525SLX2+0.75LL	103.8	0.0	0.0	-2.2	-4.1		18.9		18.6
243	1.0DL+1.0FL2+0.525SLZ1+0.75LL	106.5	0.0	0.0	-3.3	-6.0		13.3		13.1
244	1.0DL+1.0FL2+0.525SLZ2+0.75LL	102.4	0.0	0.0	-1.9	-3.6		21.6		21.3
245	0.6DL+1.0WLX1	79.7	-0.9	-1.7	-1.9	-3.5	34.7	17.3	34.3	17.0
246	0.6DL+1.0WLX2	74.6	0.9	1.7	0.8	1.5	32.5	37.8	32.1	37.3
247	0.6DL+1.0WLZ1	85.6	0.0	0.0	-4.2	-7.8		8.2		8.1
248	0.6DL+1.0WLZ2	68.7	0.0	0.0	3.1	5.8		8.9		8.8
249	0.6DL+0.6FL1+0.7SLX1	77.8	0.0	0.0	-1.0	-1.8		32.2		31.8
250	0.6DL+0.6FL1+0.7SLX2	76.0	0.0	0.0	0.0	-0.1		770.2		759.9
251	0.6DL+0.6FL1+0.7SLZ1	79.7	0.0	0.0	-1.4	-2.6		23.2		22.9
252	0.6DL+0.6FL1+0.7SLZ2	74.1	0.0	0.0	0.4	0.7		79.1		78.0
						Min.=		8.1		8.0
								OK		OK

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DESIGN OF FOUNDATION MKD. F2

Load Combination & Calculation of Base pressure

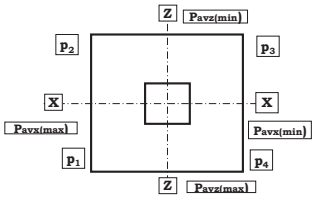
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m²

Net SBC of soil for DL, LL= 500 kN/m² For WL/SL case = 625 kN/m²

Gross SBC of soil for DL, LL= 527 kN/m² For WL/SL case = 652 kN/m²



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	Base Pressure (kN/m²) p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	Pavx(max)	Pavx(min)	Pavz(max)	Pavz(min)	Remarks
201	1.0DL+1.0FL1	82.01	0.00	-1.55	36.45	0.00	2.76	39.21	33.69	39.21	33.69	12.21	36.45	36.45	39.21	33.69	O.K
202	1.0DL+1.0FL2	81.91	0.00	-1.54	36.40	0.00	2.73	39.13	33.67	39.13	33.67	12.13	36.40	36.40	39.13	33.67	O.K
203	1.0DL+1.0FL1+1.0LL+1.0TL	112.17	0.00	-6.51	49.85	0.00	11.58	61.43	38.28	61.43	38.28	34.43	49.86	49.86	61.43	38.28	O.K
204	1.0DL+1.0FL2+1.0LL+1.0TL	112.07	0.00	-6.49	49.81	0.00	11.54	61.35	38.26	61.35	38.26	34.35	49.81	49.81	61.35	38.26	O.K
205	1.0DL+1.0FL1+0.75LL+0.75TL	104.63	0.00	-5.27	46.50	0.00	9.37	55.88	37.13	55.88	37.13	28.88	46.51	46.51	55.88	37.13	O.K
206	1.0DL+1.0FL2+0.75LL+0.75TL	104.53	0.00	-5.25	46.46	0.00	9.34	55.80	37.12	55.80	37.12	28.80	46.46	46.46	55.80	37.12	O.K
207	1.0DL+1.0FL1+1.0PL	91.09	0.00	-5.98	40.48	0.00	10.62	51.11	29.86	51.11	29.86	24.11	40.49	40.49	51.11	29.86	O.K
208	1.0DL+1.0FL2+1.0PL	90.99	0.00	-5.96	40.44	0.00	10.59	51.03	29.85	51.03	29.85	24.03	40.44	40.44	51.03	29.85	O.K
209	1.0DL+1.0FL1+1.0LL+1.0TL+1.0	121.25	0.00	-10.92	53.89	0.00	19.40	73.29	34.48	73.29	34.48	46.29	53.89	53.89	73.29	34.48	O.K
210	1.0DL+1.0FL2+1.0LL+1.0TL+1.0	121.15	0.00	-10.92	53.84	0.00	19.40	73.25	34.44	73.25	34.44	46.25	53.85	53.85	73.25	34.44	O.K
211	1.0DL+1.0FL1+0.75LL+0.75TL+1	113.71	0.00	-9.69	50.54	0.00	17.23	67.77	33.30	67.77	33.30	40.77	50.54	50.54	67.77	33.30	O.K
212	1.0DL+1.0FL2+0.75LL+0.75TL+1	113.61	0.00	-9.68	50.49	0.00	17.20	67.69	33.29	67.69	33.29	40.69	50.49	50.49	67.69	33.29	O.K
213	1.0DL+1.0FL1+1.0WLX1	84.54	-1.72	-4.03	37.57	3.06	7.17	47.80	33.46	41.68	27.34	20.80	40.63	34.51	44.74	30.40	O.K
214	1.0DL+1.0FL1+1.0WLX2	79.48	1.72	0.91	35.32	3.06	1.61	39.99	36.77	33.88	30.65	12.99	38.38	32.27	36.94	33.71	O.K
215	1.0DL+1.0FL1+1.0WLZ1	90.44	0.00	-8.36	40.20	0.00	14.87	55.06	25.33	55.06	25.33	28.06	40.20	40.20	55.06	25.33	O.K
216	1.0DL+1.0FL1+1.0WLZ2	73.58	0.00	5.24	32.70	0.00	9.31	42.01	23.39	42.01	23.39	15.01	32.70	32.70	42.01	23.39	O.K
217	1.0DL+1.0FL2+1.0WLX1	84.44	-1.72	-4.01	37.53	3.06	7.14	47.72	33.45	41.61	27.33	20.72	40.59	34.47	44.67	30.39	O.K
218	1.0DL+1.0FL2+1.0WLX2	79.38	1.72	0.93	35.28	3.06	1.64	39.98	36.69	33.87	30.58	12.98	38.34	32.23	36.93	33.64	O.K
219	1.0DL+1.0FL2+1.0WLZ1	90.34	0.00	-8.34	40.15	0.00	14.83	54.98	25.32	54.98	25.32	27.98	40.15	40.15	54.98	25.32	O.K
220	1.0DL+1.0FL2+1.0WLZ2	73.48	0.00	5.25	32.66	0.00	9.34	42.00	23.32	42.00	23.32	15.00	32.66	32.66	42.00	23.32	O.K
221	1.0DL+1.0FL1+0.7SLX1	82.92	0.00	-2.42	36.85	0.00	4.31	41.16	32.54	41.16	32.54	14.16	36.85	36.85	41.16	32.54	O.K
222	1.0DL+1.0FL1+0.7SLX2	81.09	0.02	-0.68	36.04	0.03	1.22	37.29	34.86	37.22	34.79	10.29	36.08	36.01	37.26	34.83	O.K
223	1.0DL+1.0FL1+0.7SLZ1	84.78	0.00	-3.20	37.68	0.00	5.69	43.37	31.99	43.37	31.99	16.37	37.68	37.68	43.37	31.99	O.K
224	1.0DL+1.0FL1+0.7SLZ2	79.24	0.00	0.07	35.22	0.00	0.13	35.35	35.09	35.35	35.09	8.35	35.22	35.22	35.35	35.09	O.K
225	1.0DL+1.0FL2+0.7SLX1	82.82	0.00	-2.41	36.81	0.00	4.28	41.08	32.53	41.08	32.53	14.08	36.81	36.81	41.08	32.53	O.K
226	1.0DL+1.0FL2+0.7SLX2	80.99	0.02	-0.68	36.00	0.03	1.22	37.25	34.81	37.18	34.75	10.25	36.03	35.97	37.22	34.78	O.K
227	1.0DL+1.0FL2+0.7SLZ1	84.68	0.00	-3.18	37.64	0.00	5.66	43.29	31.98	43.29	31.98	16.29	37.64	37.64	43.29	31.98	O.K
228	1.0DL+1.0FL2+0.7SLZ2	79.14	0.00	0.09	35.17	0.00	0.16	35.34	35.01	35.34	35.01	8.34	35.18	35.18	35.34	35.01	O.K
229	1.0DL+1.0FL1+0.75WLX1+0.75L	106.46	-1.28	-6.64	47.32	2.27	11.81	61.39	37.78	56.85	33.24	34.39	49.59	45.05	59.12	35.51	O.K
230	1.0DL+1.0FL1+0.75WLX2+0.75L	102.67	1.30	-2.94	45.63	2.30	5.23	53.16	42.70	48.56	38.10	26.16	47.93	43.33	50.86	40.40	O.K
231	1.0DL+1.0FL1+0.75WLZ1+0.75L	110.88	0.00	-9.90	49.28	0.00	17.60	66.88	31.68	66.88	31.68	39.88	49.28	49.28	66.88	31.68	O.K
232	1.0DL+1.0FL1+0.75WLZ2+0.75L	98.24	0.00	0.30	43.66	0.00	0.53	44.19	43.14	44.19	43.14	17.19	43.67	43.67	44.19	43.14	O.K
233	1.0DL+1.0FL2+0.75WLX1+0.75L	106.36	-1.28	-6.64	47.27	2.27	11.81	61.35	37.73	56.81	33.19	34.35	49.54	45.00	59.08	35.46	O.K
234	1.0DL+1.0FL2+0.75WLX2+0.75L	102.57	1.30	-2.92	45.59	2.30	5.20	53.09	42.69	48.48	38.09	26.09	47.89	43.29	50.79	40.39	O.K
235	1.0DL+1.0FL2+0.75WLZ1+0.75L	110.79	0.00	-9.88	49.24	0.00	17.56	66.80	31.68	66.80	31.68	39.80	49.24	49.24	66.80	31.68	O.K
236	1.0DL+1.0FL2+0.75WLZ2+0.75L	98.14	0.00	0.31	43.62	0.00	0.56	44.18	43.06	44.18	43.06	17.18	43.62	43.62	44.18	43.06	O.K
237	1.0DL+1.0FL1+0.525SLX1+0.75	105.25	0.00	-5.44	46.78	0.00	9.67	56.45	37.11	56.45	37.11	29.45	46.78	46.78	56.45	37.11	O.K

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238:1.0DL+1.0FL1+0.525SLX2+0.75	103.88	0.02	-4.14	46.17	0.03	7.37	53.57	38.83	53.50	38.77	26.57	46.20	46.14	53.54	38.80	O.K
239:1.0DL+1.0FL1+0.525SLZ1+0.75	106.64	0.00	-6.03	47.40	0.00	10.72	58.12	36.67	58.12	36.67	31.12	47.40	47.40	58.12	36.67	O.K
240:1.0DL+1.0FL1+0.525SLZ2+0.75	102.49	0.00	-3.57	45.55	0.00	6.35	51.90	39.20	51.90	39.20	24.90	45.55	45.55	51.90	39.20	O.K
241:1.0DL+1.0FL2+0.525SLX1+0.75	105.15	0.00	-5.42	46.73	0.00	9.64	56.37	37.10	56.37	37.10	29.37	46.74	46.74	56.37	37.10	O.K
242:1.0DL+1.0FL2+0.525SLX2+0.75	103.78	0.02	-4.13	46.12	0.03	7.33	53.49	38.82	53.43	38.76	26.49	46.16	46.10	53.46	38.79	O.K
243:1.0DL+1.0FL2+0.525SLZ1+0.75	106.54	0.00	-6.01	47.35	0.00	10.69	58.04	36.66	58.04	36.66	31.04	47.35	47.35	58.04	36.66	O.K
244:1.0DL+1.0FL2+0.525SLZ2+0.75	102.39	0.00	-3.55	45.51	0.00	6.31	51.82	39.19	51.82	39.19	24.82	45.51	45.51	51.82	39.19	O.K
245:0.6DL+1.0WLX1	79.70	-1.72	-3.46	35.42	3.06	6.15	44.63	32.33	38.51	26.21	17.63	38.48	32.36	41.57	29.27	O.K
246:0.6DL+1.0WLX2	74.64	1.72	1.48	33.17	3.06	2.63	38.86	33.60	32.75	27.48	11.86	36.23	30.12	35.81	30.54	O.K
247:0.6DL+1.0WLZ1	85.59	0.00	-7.79	38.04	0.00	13.85	51.89	24.19	51.89	24.19	24.89	38.04	38.04	51.89	24.19	O.K
248:0.6DL+1.0WLZ2	68.74	0.00	5.81	30.55	0.00	10.33	40.88	20.22	40.88	20.22	13.88	30.55	30.55	40.88	20.22	O.K
249:0.6DL+0.6FL1+0.7SLX1	77.82	0.00	-1.81	34.59	0.00	3.22	37.81	31.36	37.81	31.36	10.81	34.59	34.59	37.81	31.36	O.K
250:0.6DL+0.6FL1+0.7SLX2	75.99	0.02	-0.07	33.77	0.03	0.13	33.94	33.67	33.87	33.61	6.94	33.81	33.74	33.91	33.64	O.K
251:0.6DL+0.6FL1+0.7SLZ1	79.68	0.00	-2.57	35.41	0.00	4.57	39.98	30.84	39.98	30.84	12.98	35.41	35.41	39.98	30.84	O.K
252:0.6DL+0.6FL1+0.7SLZ2	74.14	0.00	0.70	32.95	0.00	1.25	34.20	31.70	34.20	31.70	7.20	32.95	32.95	34.20	31.70	O.K
Ultimate Conditions																
101:1.4DL+1.4FL1	87.11	0.00	-2.18	38.72	0.00	3.88	42.60	34.83	42.60	34.83	15.60	38.72	38.72	42.60	34.83	
102:1.4DL+1.4FL2	86.97	0.00	-2.16	38.65	0.00	3.85	42.50	34.81	42.50	34.81	15.50	38.66	38.66	42.50	34.81	
103:1.2DL+1.2FL1+1.6LL	132.68	0.00	-8.77	58.97	0.00	15.59	74.56	43.38	74.56	43.38	47.56	58.97	58.97	74.56	43.38	
104:1.2DL+1.2FL2+1.6LL	132.56	0.00	-8.75	58.92	0.00	15.56	74.47	43.36	74.47	43.36	47.47	58.92	58.92	74.47	43.36	
105:1.2DL+1.2FL1+1.2TL+1.6LL	132.78	0.00	-9.55	59.01	0.00	16.97	75.98	42.04	75.98	42.04	48.98	59.01	59.01	75.98	42.04	
106:1.2DL+1.2FL2+1.2TL+1.6LL	132.66	0.00	-9.53	58.96	0.00	16.94	75.90	42.02	75.90	42.02	48.90	58.96	58.96	75.90	42.02	
107:1.4DL+1.4FL1+1.4PL	99.82	0.00	-8.36	44.36	0.00	14.87	59.23	29.50	59.23	29.50	32.23	44.37	44.37	59.23	29.50	
108:1.4DL+1.4FL2+1.4PL	99.68	0.00	-8.33	44.30	0.00	14.80	59.10	29.50	59.10	29.50	32.10	44.30	44.30	59.10	29.50	
109:1.2DL+1.2FL1+1.6LL+1.2PL	143.57	0.00	-14.06	63.81	0.00	25.00	88.80	38.81	88.80	38.81	61.80	63.81	63.81	88.80	38.81	
110:1.2DL+1.2FL2+1.6LL+1.2PL	143.45	0.00	-14.04	63.76	0.00	24.96	88.72	38.79	88.72	38.79	61.72	63.76	63.76	88.72	38.79	
111:1.2DL+1.2FL1+1.2TL+1.6LL+1.2	143.68	0.00	-14.84	63.86	0.00	26.38	90.23	37.48	90.23	37.48	63.23	63.86	63.86	90.23	37.48	
112:1.2DL+1.2FL2+1.2TL+1.6LL+1.2	143.56	0.00	-14.82	63.80	0.00	26.34	90.15	37.46	90.15	37.46	63.15	63.81	63.81	90.15	37.46	
113:1.2DL+0.8WLX1	87.10	-1.37	-3.94	38.71	2.43	7.01	48.15	34.14	43.28	29.27	21.15	41.15	36.28	45.72	31.71	
114:1.2DL+0.8WLX2	83.05	1.37	0.00	36.91	2.43	0.00	39.34	39.34	34.48	34.48	12.34	39.34	34.48	36.91	36.91	
115:1.2DL+0.8WLZ1	91.82	0.00	-7.40	40.81	0.00	13.16	53.96	27.65	53.96	27.65	26.96	40.81	40.81	53.96	27.65	
116:1.2DL+0.8WLZ2	78.33	0.00	3.46	34.81	0.00	6.15	40.96	28.66	40.96	28.66	13.96	34.81	34.81	40.96	28.66	
117:1.2DL+1.6WLX1+0.5LL	104.16	-2.74	-8.08	46.29	4.87	14.37	65.53	36.79	55.80	27.05	38.53	51.16	41.43	60.67	31.92	
118:1.2DL+1.6WLX2+0.5LL	96.06	2.74	-0.17	42.69	4.87	0.30	47.86	47.26	38.12	37.53	20.86	47.56	37.83	42.99	42.40	
119:1.2DL+1.6WLZ1+0.5LL	113.60	0.00	-15.00	50.49	0.00	26.67	77.16	23.82	77.16	23.82	50.16	50.49	50.49	77.16	23.82	
120:1.2DL+1.6WLZ2+0.5LL	86.63	0.00	6.75	38.50	0.00	12.00	50.51	26.50	50.51	26.50	23.51	38.51	38.51	50.51	26.50	
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	101.20	-0.02	-5.33	44.98	0.03	9.47	54.48	35.54	54.42	35.47	27.48	45.01	44.95	54.45	35.51	
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	98.59	0.02	-2.85	43.82	0.03	5.06	48.92	38.79	48.85	38.72	21.92	43.86	43.79	48.89	38.76	
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	103.86	0.00	-6.44	46.16	0.00	11.45	57.61	34.71	57.61	34.71	30.61	46.16	46.16	57.61	34.71	
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	95.94	0.00	-1.74	42.64	0.00	3.09	45.73	39.55	45.73	39.55	18.73	42.64	42.64	45.73	39.55	
125:0.9DL+1.6WLX1	85.17	-2.74	-5.44	37.85	4.87	9.67	52.39	33.05	42.66	23.32	25.39	42.72	32.99	47.53	28.19	
126:0.9DL+1.6WLX2	77.07	2.74	2.48	34.25	4.87	4.41	43.53	34.71	33.79	24.98	16.53	39.12	29.39	38.66	29.85	
127:0.9DL+1.6WLZ1	94.60	0.00	-12.36	42.04	0.00	21.97	64.01	20.07	64.01	20.07	37.01	42.04	42.04	64.01	20.07	
128:0.9DL+1.6WLZ2	67.64	0.00	9.40	30.06	0.00	16.71	46.77	13.35	46.77	13.35	19.77	30.06	30.06	46.77	13.35	
129:0.9DL+1.0SLX1	82.43	-0.02	-2.72	36.64	0.03	4.83	41.50	31.83	41.44	31.77	14.50	36.67	36.61	41.47	31.80	
130:0.9DL+1.0SLX2	79.81	0.02	-0.24	35.47	0.03	0.43	35.93	35.08	35.87	35.01	8.93	35.51	35.44	35.90	35.05	
131:0.9DL+1.0SLZ1	85.08	0.00	-3.83	37.81	0.00	6.81	44.62	31.01	44.62	31.01	17.62	37.82	37.82	44.62	31.01	
132:0.9DL+1.0SLZ2	77.16	0.00	0.87	34.29	0.00	1.55	35.84	32.75	35.84	32.75	8.84	34.30	34.30	35.84	32.75	
Max :	144	3	9													
Min :	68	-3	-15													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 12.21 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 22.21 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 46.29 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 56.29 kN/m²

Sample calculation of values of base pressure:

Load case : 202:1.0DL+1.0FL2

Maximum pressure, p_1 (pmax) = 36.40+0.00+2.73= 39.13 kN/m²

p_2= 36.40+0.00-2.73= 33.67 kN/m²

p_3= 36.40-(0.00+2.73)= 33.67 kN/m²

p_4= 36.40-0.00+2.73= 39.13 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 39.13-1.5*18= 12.13 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (39.13+33.67)/2= 36.40 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (33.67+39.13)/2= 36.40 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (39.13+39.13)/2= 39.13 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (33.67+33.67)/2= 33.67 kN/m²

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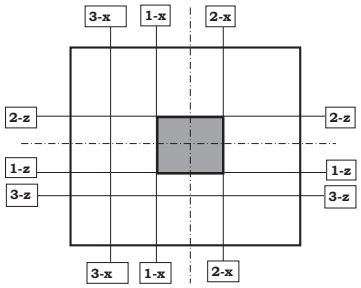
DESIGN OF FOUNDATION MKD. F2

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.
p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.
p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.
BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.
BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.
SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.45+2*0.5*0.282)+(0.55+2*0.5*0.282))*2
= 3.128 m
Punching Area= ((0.45+2*0.5*0.282)*(0.55+2*0.5*0.282))
= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.35*24+1.15*18)
= 29.10 kN/m^2
Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL+1.4FL1	38.72	38.72	38.72	1.33	1.33	2.22	15.77
102	1.4DL+1.4FL2	38.66	38.66	38.66	1.32	1.32	2.21	15.67
103	1.2DL+1.2FL1+1.6LL	58.97	58.97	58.97	4.12	4.12	6.90	48.98
104	1.2DL+1.2FL2+1.6LL	58.92	58.92	58.92	4.11	4.11	6.89	48.90
105	1.2DL+1.2FL1+1.2TL+1.6LL	59.01	59.01	59.01	4.12	4.12	6.91	49.06
106	1.2DL+1.2FL2+1.2TL+1.6LL	58.96	58.96	58.96	4.12	4.12	6.90	48.97
107	1.4DL+1.4FL1+1.4PL	44.37	44.37	44.37	2.10	2.10	3.53	25.03
108	1.4DL+1.4FL2+1.4PL	44.30	44.30	44.30	2.09	2.09	3.51	24.93
109	1.2DL+1.2FL1+1.6LL+1.2PL	63.81	63.81	63.81	4.78	4.78	8.02	56.92
110	1.2DL+1.2FL2+1.6LL+1.2PL	63.76	63.76	63.76	4.78	4.78	8.01	56.84
111	1.2DL+1.2FL1+1.2TL+1.6LL+1	63.86	63.86	63.86	4.79	4.79	8.03	57.00
112	1.2DL+1.2FL2+1.2TL+1.6LL+1	63.81	63.81	63.81	4.78	4.78	8.02	56.92
113	1.2DL+0.8WLX1	39.45	37.98	40.40	1.58	1.07	2.70	15.76
114	1.2DL+0.8WLX2	37.64	36.18	38.59	1.33	0.82	2.28	12.81
115	1.2DL+0.8WLZ1	40.81	40.81	40.81	1.61	1.61	2.71	19.20
116	1.2DL+0.8WLZ2	34.81	34.81	34.81	0.79	0.79	1.32	9.37
117	1.2DL+1.6WLX1+0.5LL	47.75	44.84	49.66	2.88	1.86	4.92	28.20
118	1.2DL+1.6WLX2+0.5LL	44.15	41.24	46.06	2.39	1.36	4.09	22.29
119	1.2DL+1.6WLZ1+0.5LL	50.49	50.49	50.49	2.95	2.95	4.94	35.08
120	1.2DL+1.6WLZ2+0.5LL	38.51	38.51	38.51	1.30	1.30	2.17	15.42
121	1.2DL+1.0SLX1+0.5LL+0.5FL1	44.99	44.97	45.00	2.19	2.19	3.67	26.04
122	1.2DL+1.0SLX2+0.5LL+0.5FL1	43.84	43.81	43.85	2.03	2.03	3.41	24.14
123	1.2DL+1.0SLZ1+0.5LL+0.5FL1	46.16	46.16	46.16	2.35	2.35	3.94	27.98
124	1.2DL+1.0SLZ2+0.5LL+0.5FL1	42.64	42.64	42.64	1.87	1.87	3.13	22.21
125	0.9DL+1.6WLX1	39.31	36.40	41.22	1.72	0.69	2.97	14.36
126	0.9DL+1.6WLX2	35.71	32.80	37.62	1.22	0.20	2.14	8.45
127	0.9DL+1.6WLZ1	42.04	42.04	42.04	1.78	1.78	2.99	21.23
128	0.9DL+1.6WLZ2	30.06	30.06	30.06	0.13	0.13	0.22	1.58
129	0.9DL+1.0SLX1	36.65	36.63	36.66	1.04	1.04	1.75	12.36
130	0.9DL+1.0SLX2	35.49	35.46	35.50	0.88	0.87	1.48	10.45
131	0.9DL+1.0SLZ1	37.82	37.82	37.82	1.20	1.20	2.01	14.29
132	0.9DL+1.0SLZ2	34.30	34.30	34.30	0.72	0.72	1.20	8.52

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Sample calculation of BM & SF

Load case : 101:1.4DL+1.4FL1

p_{1x} =	38.72-(38.72-38.72)*0.525/1.5 =	38.72 kN/m^2	Overhang =	0.525 m
p_{2x} =	38.72+(38.72-38.72)*0.525/1.5 =	38.72 kN/m^2	Effective depth =	0.294 m
p_{3x} =	38.72-(38.72-38.72)*(0.525-0.294)/1.5 =	38.72 kN/m^2		
BM_{1x} =	(38.72-29.1)*0.525*0.525/2+0.5*(38.72-38.72)*0.525*0.525*0.67 =	1.33 kN-m/m		
BM_{2x} =	(38.72-29.1)*0.525*0.525/2-0.5*(-38.72+38.72)*0.525*0.525*0.67 =	1.33 kN-m/m		
SF_{3x} =	(38.72-29.1)*(0.525-0.294)+0.5*(38.72-38.72)*(0.525-0.294) =	2.22 kN/m		
$S.F_{(punching)}$ =	(2.25-0.61)*(38.72-29.1) =	15.77 kN		

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	4.79	0.06	0.07	0.000	51.79	529.20	12ϕT	213.71	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 8.03 kN/m

Shear stress , τ_v= (8.03*1000)/(1000*294)

= 0.03 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 57.00 kN

Strength of Concrete in shear Vc1 = 3123.32 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 1759.04 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 1813.54 kN **Table 22.6.5.2_ (c)**

Design Vc= 1759.04 kN

Allowable shear stress, (ϕ Vc)/bd = 1.43 N/mm^2

Design punching shear stress, τ = (57*1000)/(3128*294)

= 0.06 N/mm^2 **No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in kN/m^2.
p2z =pressure at the face of column at section 2z-2z in kN/m^2.
p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	B.M_{1z} kNm	B.M_{2z} kNm	S.F_{3z} kN
Ultimate Conditions							
101:1.4DL+1.4FL1		40.14	37.29	41.60	1.43	0.74	2.51
102:1.4DL+1.4FL2		40.06	39.97	41.51	1.42	0.46	2.49
103:1.2DL+1.2FL1+1.6LL		64.69	45.80	70.55	4.76	0.68	8.39
104:1.2DL+1.2FL2+1.6LL		64.62	52.84	70.47	4.75	1.29	8.37
105:1.2DL+1.2FL1+1.2TL+1.6LL		65.23	51.92	71.61	4.89	1.25	8.63
106:1.2DL+1.2FL2+1.2TL+1.6LL		65.17	50.17	71.54	4.88	1.10	8.61
107:1.4DL+1.4FL1+1.4PL		49.82	41.59	55.40	3.05	0.78	5.45
108:1.4DL+1.4FL2+1.4PL		49.73	34.33	55.29	3.04	0.29	5.42
109:1.2DL+1.2FL1+1.6LL+1.2PL		72.97	38.22	82.37	6.15	0.69	10.90
110:1.2DL+1.2FL2+1.6LL+1.2PL		72.91	43.94	82.30	6.14	0.27	10.89
111:1.2DL+1.2FL1+1.2TL+1.6LL+1		73.53	38.91	83.44	6.27	-0.07	11.14
112:1.2DL+1.2FL2+1.2TL+1.6LL+1		73.46	39.74	83.37	6.27	0.49	11.13
113:1.2DL+0.8WLX1		41.28	33.36	43.92	1.71	0.50	3.03
114:1.2DL+0.8WLX2		36.91	42.63	36.91	0.88	0.22	1.51
115:1.2DL+0.8WLZ1		45.63	32.19	50.57	2.49	-0.32	4.47

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116:1.2DL+0.8WLZ2	37.07	32.62	39.38	1.19	0.39	2.14
117:1.2DL+1.6WLX1+0.5LL	51.57	33.61	56.97	3.22	0.60	5.74
118:1.2DL+1.6WLX2+0.5LL	42.80	42.71	42.91	1.56	0.73	2.67
119:1.2DL+1.6WLZ1+0.5LL	60.27	27.47	70.30	4.79	0.39	8.61
120:1.2DL+1.6WLZ2+0.5LL	42.91	31.12	47.42	2.13	0.33	3.83
121:1.2DL+1.0SLX1+0.5LL+0.5FL1	48.45	36.05	52.01	2.64	0.69	4.66
122:1.2DL+1.0SLX2+0.5LL+0.5FL1	45.68	40.19	47.59	2.11	0.63	3.69
123:1.2DL+1.0SLZ1+0.5LL+0.5FL1	50.36	34.91	54.66	2.95	0.66	5.22
124:1.2DL+1.0SLZ2+0.5LL+0.5FL1	43.77	45.75	44.93	1.80	1.10	3.13
125:0.9DL+1.6WLX1	41.41	35.37	45.04	1.85	1.09	3.32
126:0.9DL+1.6WLX2	35.87	41.58	37.53	0.97	0.66	1.74
127:0.9DL+1.6WLZ1	50.10	27.71	58.36	3.42	1.01	6.19
128:0.9DL+1.6WLZ2	36.19	27.83	42.47	1.60	0.43	3.00
129:0.9DL+1.0SLX1	38.41	37.81	40.23	1.28	1.18	2.27
130:0.9DL+1.0SLX2	35.63	45.10	35.79	0.76	0.79	1.30
131:0.9DL+1.0SLZ1	40.31	35.18	42.87	1.59	1.28	2.83
132:0.9DL+1.0SLZ2	34.86	40.26	35.44	0.72	1.02	1.26

Sample calculation of BM & SF

Load case : 101:1.4DL+1.4FL1

p_{1z} = 42.6-(42.6-34.83)*0.475/1.5= 40.14 kN/m^2

p_{2z} = 34.83+(42.6-34.83)*0.475/1.5= 37.29 kN/m^2 Overhang = 0.475 m

p_{3z} = 42.6-(42.6-34.83)*(0.475-0.282)/1.5= 41.60 kN/m^2 Effective depth = 0.282 m

BM_{1z} = (40.14-29.1)*0.475*0.475/2+0.5*(42.6-40.14)*0.475*0.475*0.67 = 1.43 kN-m/m

BM_{2z} = (37.29-29.1)*0.475*0.475/2-0.5*(-34.83+37.29)*0.475*0.475*0.67 = 0.74 kN-m/m

SF_{3z} = (41.60-29.1)*(0.475-0.282)+0.5*(42.6-41.60)*(0.475-0.282) = 2.51 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	6.27	0.079	0.105	0.0003	73.77	507.60	12ϕT	222.81	300	200	565.5

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 11.14 kN/m

Shear stress , τ_v= (11.14*1000)/(1000*282)

= 0.04 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **200** c/c

At Top **10ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **200** c/c

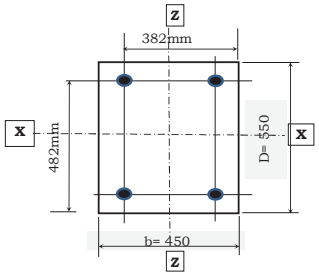
At Top **10ϕT** **200** c/c

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DESIGN OF COLUMN MKD **C2**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	16mm	
Dia. of link=	10mm	
Effective cover =	68mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupprotted length of column about X (l_{ux}) =	1450mm
Unsupprotted length of column about Z (l_{uz}) =	1450mm

For Columns not braced against sidesway,

Where, Effective length factor K= 1.2

Slenderness ratio,	$K l_{ux} / r_x$ =	10.96	< 22 SECTION 6.2.5a	Radius of gyration r_x= 159 mm
	$K l_{uz} / r_z$ =	13.40	< 22 SECTION 6.2.5a	Radius of gyration r_z= 130 mm

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γ_y = 0.79

γ_z = 0.74

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = Pn/f'_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
101	1.4DL+1.4FL1	1.79	0.0	0.2	2.7	0.0	0.3	0.003
102	1.4DL+1.4FL2	1.77	0.0	0.2	2.7	0.0	0.3	0.003
103	1.2DL+1.2FL1+1.6LL	6.34	0.0	0.7	9.8	0.0	1.1	0.011
104	1.2DL+1.2FL2+1.6LL	6.33	0.0	0.7	9.7	0.0	1.1	0.011
105	1.2DL+1.2FL1+1.2TL+1.6LL	6.35	0.0	0.8	9.8	0.0	1.2	0.011
106	1.2DL+1.2FL2+1.2TL+1.6LL	6.34	0.0	0.8	9.8	0.0	1.2	0.011
107	1.4DL+1.4FL1+1.4PL	3.06	0.0	0.7	4.7	0.0	1.0	0.005
108	1.4DL+1.4FL2+1.4PL	3.04	0.0	0.7	4.7	0.0	1.0	0.005
109	1.2DL+1.2FL1+1.6LL+1.2PL	7.43	0.0	1.1	11.4	0.0	1.8	0.013
110	1.2DL+1.2FL2+1.6LL+1.2PL	7.42	0.0	1.1	11.4	0.0	1.8	0.013
111	1.2DL+1.2FL1+1.2TL+1.6LL+1.2PL	7.44	0.0	1.2	11.4	0.0	1.9	0.013
112	1.2DL+1.2FL2+1.2TL+1.6LL+1.2PL	7.43	0.0	1.2	11.4	0.0	1.8	0.013
113	1.2DL+0.8WLX1	1.78	0.1	0.3	2.7	0.2	0.5	0.003
114	1.2DL+0.8WLX2	1.38	0.1	0.0	2.1	0.2	0.0	0.002
115	1.2DL+0.8WLZ1	2.26	0.0	0.6	3.5	0.0	0.9	0.004
116	1.2DL+0.8WLZ2	0.91	0.0	0.3	1.4	0.0	0.4	0.002
117	1.2DL+1.6WLX1+0.5LL	3.49	0.2	0.7	5.4	0.3	1.0	0.006
118	1.2DL+1.6WLX2+0.5LL	2.68	0.2	0.0	4.1	0.3	0.0	0.005
119	1.2DL+1.6WLZ1+0.5LL	4.43	0.0	1.2	6.8	0.0	1.9	0.008
120	1.2DL+1.6WLZ2+0.5LL	1.74	0.0	0.5	2.7	0.0	0.8	0.003
121	1.2DL+1.0SLX1+0.5LL+0.5FL1	3.19	0.0	0.4	4.9	0.0	0.7	0.006
122	1.2DL+1.0SLX2+0.5LL+0.5FL1	2.93	0.0	0.2	4.5	0.0	0.4	0.005
123	1.2DL+1.0SLZ1+0.5LL+0.5FL1	3.46	0.0	0.5	5.3	0.0	0.8	0.006
124	1.2DL+1.0SLZ2+0.5LL+0.5FL1	2.67	0.0	0.1	4.1	0.0	0.2	0.005
125	0.9DL+1.6WLX1	1.59	0.2	0.4	2.4	0.3	0.7	0.003
126	0.9DL+1.6WLX2	0.78	0.2	0.2	1.2	0.3	0.3	0.001
127	0.9DL+1.6WLZ1	2.53	0.0	1.0	3.9	0.0	1.5	0.005
128	0.9DL+1.6WLZ2	-0.16	0.0	0.8	-0.2	0.0	1.2	0.000
129	0.9DL+1.0SLX1	1.32	0.0	0.2	2.0	0.0	0.3	0.002
130	0.9DL+1.0SLX2	1.06	0.0	0.0	1.6	0.0	0.0	0.002
131	0.9DL+1.0SLZ1	1.58	0.0	0.3	2.4	0.0	0.5	0.003
132	0.9DL+1.0SLZ2	0.79	0.0	0.1	1.2	0.0	0.1	0.001

Reinforcement: Provide,

4-Nos	16 mm Dia	804.57 Sq mm	(At Corner)
4-Nos	16 mm Dia	804.57 Sq mm	(Along Z axis)
3-Nos	16 mm Dia	603.43 Sq mm	(Along X axis)

A_{sc} pvd. = 2212.57 Sq mm

Total A_{sc} = 0.89 %

β = P_n / P_0 (range is 0.55 to 0.70)

Where, P_0 = 0.85 f'_c (A_g - A_{sc}) + f_y A_{sc} = 822.6581 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f'_cb^2D$	$R_{nx}=M_{nox}/f'_cb^2D$						
101	0.05	0.05	19.5	23.8	0.55	1.16	0.0056	O.K
102	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
103	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
104	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
105	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
106	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
107	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
108	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
109	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
110	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
111	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
112	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
113	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K
114	0.05	0.05	19.5	23.8	0.55	1.16	0.00	O.K
115	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K
116	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
117	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
118	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
119	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
120	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K
121	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
122	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
123	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
124	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
125	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K
126	0.05	0.05	19.5	23.8	0.55	1.16	0.00	O.K
127	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K
128	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
129	0.05	0.05	19.5	23.8	0.55	1.16	0.03	O.K
130	0.05	0.05	19.5	23.8	0.55	1.16	0.01	O.K
131	0.05	0.05	19.5	23.8	0.55	1.16	0.05	O.K
132	0.05	0.05	19.5	23.8	0.55	1.16	0.02	O.K

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 BEAM FORCE DETAIL

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Beam Force Detail

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
1	100:1.4 (D + F	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	1.168	0.000	0.000	0.000	-0.018
		0.055	0.000	2.334	0.000	0.000	0.000	-0.067
		0.082	0.000	3.498	0.000	0.000	0.000	-0.147
		0.110	0.000	4.661	0.000	0.000	0.000	-0.258
		0.137	0.000	5.821	0.000	0.000	0.000	-0.401
		0.165	0.000	6.979	0.000	0.000	0.000	-0.579
		0.192	0.000	8.134	0.000	0.000	0.000	-0.787
		0.220	0.000	9.288	0.000	0.000	0.000	-1.027
		0.247	0.000	10.440	0.000	0.000	0.000	-1.297
		0.275	-0.000	11.590	-0.000	-0.000	-0.000	-1.598
	101:1.2(D + F'	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		0.027	0.000	0.910	0.000	0.000	0.000	-0.014
		0.055	0.000	1.818	0.000	0.000	0.000	-0.052
		0.082	0.000	2.725	0.000	0.000	0.000	-0.115
		0.110	0.000	3.630	0.000	0.000	0.000	-0.201
		0.137	0.000	4.533	0.000	0.000	0.000	-0.312
		0.165	0.000	5.435	0.000	0.000	0.000	-0.451
		0.192	0.000	6.334	0.000	0.000	0.000	-0.613
		0.220	0.000	7.232	0.000	0.000	0.000	-0.800
		0.247	0.000	8.128	0.000	0.000	0.000	-1.010
		0.275	-0.000	9.023	-0.000	-0.000	-0.000	-1.245
	102:1.2(D + F'	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		0.027	0.000	1.104	0.000	0.000	0.000	-0.017
		0.055	0.000	2.209	0.000	0.000	0.000	-0.063
		0.082	0.000	3.313	0.000	0.000	0.000	-0.139
		0.110	0.000	4.418	0.000	0.000	0.000	-0.245
		0.137	0.000	5.522	0.000	0.000	0.000	-0.380
		0.165	0.000	6.626	0.000	0.000	0.000	-0.548
		0.192	0.000	7.731	0.000	0.000	0.000	-0.747
		0.220	0.000	8.835	0.000	0.000	0.000	-0.974
		0.247	0.000	9.940	0.000	0.000	0.000	-1.232
		0.275	-0.000	11.044	-0.000	-0.000	-0.000	-1.519
	103:1.2(D + F2	0.000	0.000	-0.000	0.000	0.000	0.000	0.000
		0.027	0.000	0.914	0.000	0.000	0.000	-0.014
		0.055	0.000	1.827	0.000	0.000	0.000	-0.052
		0.082	0.000	2.737	0.000	0.000	0.000	-0.115
		0.110	0.000	3.646	0.000	0.000	0.000	-0.202
		0.137	0.000	4.554	0.000	0.000	0.000	-0.314
		0.165	0.000	5.459	0.000	0.000	0.000	-0.453
		0.192	0.000	6.362	0.000	0.000	0.000	-0.616
		0.220	0.000	7.264	0.000	0.000	0.000	-0.803
		0.247	0.000	8.164	0.000	0.000	0.000	-1.015
		0.275	-0.000	9.063	-0.000	-0.000	-0.000	-1.250
	104:1.2(D + F2	0.000	0.000	-0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	0.913	0.000	0.000	0.000	-0.014

Beam Force Detail Cont...

Beam	L/C	d	Axial	Shear		Torsion	Bending	
			Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.055	0.000	1.824	0.000	0.000	0.000	-0.052
		0.082	0.000	2.733	0.000	0.000	0.000	-0.115
		0.110	0.000	3.641	0.000	0.000	0.000	-0.202
		0.137	0.000	4.547	0.000	0.000	0.000	-0.313
		0.165	0.000	5.452	0.000	0.000	0.000	-0.452
		0.192	0.000	6.354	0.000	0.000	0.000	-0.615
		0.220	0.000	7.255	0.000	0.000	0.000	-0.802
		0.247	0.000	8.155	0.000	0.000	0.000	-1.013
		0.275	-0.000	9.053	-0.000	-0.000	-0.000	-1.249
	105:0.9 D + 1.6	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	0.537	0.000	0.000	0.000	-0.008
		0.055	0.000	1.072	0.000	0.000	0.000	-0.031
		0.082	0.000	1.606	0.000	0.000	0.000	-0.068
		0.110	0.000	2.138	0.000	0.000	0.000	-0.119
		0.137	0.000	2.669	0.000	0.000	0.000	-0.184
		0.165	0.000	3.198	0.000	0.000	0.000	-0.265
		0.192	0.000	3.726	0.000	0.000	0.000	-0.361
		0.220	0.000	4.252	0.000	0.000	0.000	-0.471
		0.247	0.000	4.776	0.000	0.000	0.000	-0.594
		0.275	-0.000	5.299	-0.000	-0.000	-0.000	-0.732
	106:0.9 D + 1.6	0.000	0.000	-0.000	0.000	0.000	0.000	0.000
		0.027	0.000	0.770	0.000	0.000	0.000	-0.012
		0.055	0.000	1.536	0.000	0.000	0.000	-0.044
		0.082	0.000	2.298	0.000	0.000	0.000	-0.097
		0.110	0.000	3.056	0.000	0.000	0.000	-0.170
		0.137	0.000	3.811	0.000	0.000	0.000	-0.263
		0.165	0.000	4.561	0.000	0.000	0.000	-0.379
		0.192	0.000	5.307	0.000	0.000	0.000	-0.516
		0.220	0.000	6.050	0.000	0.000	0.000	-0.672
		0.247	0.000	6.789	0.000	0.000	0.000	-0.848
		0.275	-0.000	7.524	-0.000	-0.000	-0.000	-1.043
	107:0.9 (D+ F2	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	0.523	0.000	0.000	0.000	-0.008
		0.055	0.000	1.045	0.000	0.000	0.000	-0.030
		0.082	0.000	1.566	0.000	0.000	0.000	-0.066
		0.110	0.000	2.085	0.000	0.000	0.000	-0.116
		0.137	0.000	2.603	0.000	0.000	0.000	-0.179
		0.165	0.000	3.119	0.000	0.000	0.000	-0.259
		0.192	0.000	3.635	0.000	0.000	0.000	-0.352
		0.220	0.000	4.149	0.000	0.000	0.000	-0.459
		0.247	0.000	4.661	0.000	0.000	0.000	-0.580
		0.275	-0.000	5.173	-0.000	-0.000	-0.000	-0.714
	108:0.9 (D+ F2	0.000	0.000	-0.000	0.000	0.000	0.000	0.000
		0.027	0.000	0.545	0.000	0.000	0.000	-0.008
		0.055	0.000	1.089	0.000	0.000	0.000	-0.031
		0.082	0.000	1.631	0.000	0.000	0.000	-0.069
		0.110	0.000	2.172	0.000	0.000	0.000	-0.120
		0.137	0.000	2.712	0.000	0.000	0.000	-0.187

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.165	0.000	3.249	0.000	0.000	0.000	-0.270
		0.192	0.000	3.786	0.000	0.000	0.000	-0.367
		0.220	0.000	4.321	0.000	0.000	0.000	-0.478
		0.247	0.000	4.855	0.000	0.000	0.000	-0.604
		0.275	-0.000	5.387	-0.000	-0.000	-0.000	-0.744
	109:1.2DL+1.2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		0.027	0.000	2.441	0.000	0.000	0.000	-0.037
		0.055	0.000	4.874	0.000	0.000	0.000	-0.140
		0.082	0.000	7.298	0.000	0.000	0.000	-0.307
		0.110	0.000	9.714	0.000	0.000	0.000	-0.539
		0.137	0.000	12.121	0.000	0.000	0.000	-0.836
		0.165	0.000	14.518	0.000	0.000	0.000	-1.206
		0.192	0.000	16.907	0.000	0.000	0.000	-1.640
		0.220	0.000	19.287	0.000	0.000	0.000	-2.137
		0.247	0.000	21.658	0.000	0.000	0.000	-2.699
		0.275	-0.000	24.021	-0.000	-0.000	-0.000	-3.323
	110:1.2DL+1.6\	0.000	0.000	-0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	1.368	0.000	0.000	0.000	-0.021
		0.055	0.000	2.732	0.000	0.000	0.000	-0.078
		0.082	0.000	4.092	0.000	0.000	0.000	-0.172
		0.110	0.000	5.449	0.000	0.000	0.000	-0.302
		0.137	0.000	6.801	0.000	0.000	0.000	-0.469
		0.165	0.000	8.150	0.000	0.000	0.000	-0.676
		0.192	0.000	9.494	0.000	0.000	0.000	-0.920
		0.220	0.000	10.835	0.000	0.000	0.000	-1.200
		0.247	0.000	12.172	0.000	0.000	0.000	-1.515
		0.275	-0.000	13.505	-0.000	-0.000	-0.000	-1.866
	111:1.2DL+1.6\	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		0.027	0.000	1.601	0.000	0.000	0.000	-0.024
		0.055	0.000	3.196	0.000	0.000	0.000	-0.092
		0.082	0.000	4.785	0.000	0.000	0.000	-0.201
		0.110	0.000	6.368	0.000	0.000	0.000	-0.354
		0.137	0.000	7.945	0.000	0.000	0.000	-0.548
		0.165	0.000	9.515	0.000	0.000	0.000	-0.790
		0.192	0.000	11.078	0.000	0.000	0.000	-1.075
		0.220	0.000	12.636	0.000	0.000	0.000	-1.401
		0.247	0.000	14.187	0.000	0.000	0.000	-1.769
		0.275	-0.000	15.733	-0.000	-0.000	-0.000	-2.178
	112:1.2DL+1.0\	0.000	0.000	-0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	1.376	0.000	0.000	0.000	-0.021
		0.055	0.000	2.748	0.000	0.000	0.000	-0.079
		0.082	0.000	4.116	0.000	0.000	0.000	-0.173
		0.110	0.000	5.480	0.000	0.000	0.000	-0.304
		0.137	0.000	6.841	0.000	0.000	0.000	-0.471
		0.165	0.000	8.197	0.000	0.000	0.000	-0.680
		0.192	0.000	9.550	0.000	0.000	0.000	-0.925
		0.220	0.000	10.898	0.000	0.000	0.000	-1.207
		0.247	0.000	12.243	0.000	0.000	0.000	-1.524

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.275	-0.000	13.584	-0.000	-0.000	-0.000	-1.877
	113:1.2DL+1.0	0.000	0.000	-0.000	0.000	0.000	0.000	-0.000
		0.027	0.000	1.398	0.000	0.000	0.000	-0.021
		0.055	0.000	2.791	0.000	0.000	0.000	-0.080
		0.082	0.000	4.181	0.000	0.000	0.000	-0.176
		0.110	0.000	5.567	0.000	0.000	0.000	-0.309
		0.137	0.000	6.949	0.000	0.000	0.000	-0.479
		0.165	0.000	8.326	0.000	0.000	0.000	-0.691
		0.192	0.000	9.700	0.000	0.000	0.000	-0.940
		0.220	0.000	11.069	0.000	0.000	0.000	-1.225
		0.247	0.000	12.435	0.000	0.000	0.000	-1.548
		0.275	-0.000	13.797	-0.000	-0.000	-0.000	-1.906
2	100:1.4 (D + F	0.000	0.000	-103.643	0.000	0.000	0.000	12.022
		0.580	0.000	-79.942	0.000	0.000	0.000	64.412
		1.159	0.000	-57.124	0.000	0.000	0.000	103.713
		1.739	0.000	-35.190	0.000	0.000	0.000	130.412
		2.318	0.000	-14.140	0.000	0.000	0.000	144.998
		2.898	0.000	6.028	0.000	0.000	0.000	147.957
		3.477	0.000	25.185	0.000	0.000	0.000	138.255
		4.057	0.000	43.459	0.000	0.000	0.000	118.025
		4.636	0.000	60.850	0.000	0.000	0.000	87.752
		5.216	0.000	77.358	0.000	0.000	0.000	47.926
		5.795	-0.000	92.982	-0.000	-0.000	-0.000	-0.968
	101:1.2(D + F	0.000	0.000	-79.214	0.000	0.000	0.000	10.455
		0.580	0.000	-60.819	0.000	0.000	0.000	50.403
		1.159	0.000	-43.183	0.000	0.000	0.000	80.214
		1.739	0.000	-26.305	0.000	0.000	0.000	100.306
		2.318	0.000	-10.186	0.000	0.000	0.000	111.099
		2.898	0.000	5.176	0.000	0.000	0.000	113.010
		3.477	0.000	19.670	0.000	0.000	0.000	105.309
		4.057	0.000	33.406	0.000	0.000	0.000	89.669
		4.636	0.000	46.384	0.000	0.000	0.000	66.508
		5.216	0.000	58.604	0.000	0.000	0.000	36.244
		5.795	-0.000	70.065	-0.000	-0.000	-0.000	-0.703
	102:1.2(D + F	0.000	0.000	-90.619	0.000	0.000	0.000	-28.519
		0.580	0.000	-68.367	0.000	0.000	0.000	16.749
		1.159	0.000	-47.901	0.000	0.000	0.000	50.014
		1.739	0.000	-29.221	0.000	0.000	0.000	72.261
		2.318	0.000	-12.327	0.000	0.000	0.000	84.477
		2.898	0.000	2.782	0.000	0.000	0.000	87.647
		3.477	0.000	15.849	0.000	0.000	0.000	81.746
		4.057	0.000	27.130	0.000	0.000	0.000	69.017
		4.636	0.000	36.625	0.000	0.000	0.000	50.446
		5.216	0.000	44.334	0.000	0.000	0.000	27.017
		5.795	-0.000	50.257	-0.000	-0.000	-0.000	-0.282
	103:1.2(D + F	0.000	0.000	-83.512	0.000	0.000	0.000	-13.050
		0.580	0.000	-65.046	0.000	0.000	0.000	29.365
		1.159	0.000	-47.347	0.000	0.000	0.000	61.605

Beam Force Detail Cont...

Beam	L/C	d	Axial	Shear		Torsion	Bending	
			Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		1.739	0.000	-30.414	0.000	0.000	0.000	84.094
		2.318	0.000	-14.247	0.000	0.000	0.000	97.255
		2.898	0.000	1.153	0.000	0.000	0.000	101.509
		3.477	0.000	15.678	0.000	0.000	0.000	96.130
		4.057	0.000	29.437	0.000	0.000	0.000	82.796
		4.636	0.000	42.429	0.000	0.000	0.000	61.930
		5.216	0.000	54.655	0.000	0.000	0.000	33.956
		5.795	-0.000	66.115	-0.000	-0.000	-0.000	-0.703
	104:1.2(D + F	0.000	0.000	-84.256	0.000	0.000	0.000	-12.249
		0.580	0.000	-65.761	0.000	0.000	0.000	30.590
		1.159	0.000	-47.980	0.000	0.000	0.000	63.223
		1.739	0.000	-30.914	0.000	0.000	0.000	86.043
		2.318	0.000	-14.562	0.000	0.000	0.000	99.445
		2.898	0.000	1.076	0.000	0.000	0.000	103.823
		3.477	0.000	15.897	0.000	0.000	0.000	98.396
		4.057	0.000	30.003	0.000	0.000	0.000	84.831
		4.636	0.000	43.395	0.000	0.000	0.000	63.525
		5.216	0.000	56.073	0.000	0.000	0.000	34.870
		5.795	-0.000	68.037	-0.000	-0.000	-0.000	-0.739
	105:0.9 D + 1.(	0.000	0.000	-41.958	0.000	0.000	0.000	9.468
		0.580	0.000	-31.333	0.000	0.000	0.000	30.331
		1.159	0.000	-21.372	0.000	0.000	0.000	45.407
		1.739	0.000	-12.075	0.000	0.000	0.000	55.062
		2.318	0.000	-3.443	0.000	0.000	0.000	59.662
		2.898	0.000	4.525	0.000	0.000	0.000	59.574
		3.477	0.000	11.734	0.000	0.000	0.000	54.600
		4.057	0.000	18.279	0.000	0.000	0.000	45.764
		4.636	0.000	24.159	0.000	0.000	0.000	33.431
		5.216	0.000	29.376	0.000	0.000	0.000	17.968
		5.795	-0.000	33.928	-0.000	-0.000	-0.000	-0.258
	106:0.9 D + 1.(	0.000	0.000	-43.955	0.000	0.000	0.000	24.957
		0.580	0.000	-29.478	0.000	0.000	0.000	45.690
		1.159	0.000	-16.691	0.000	0.000	0.000	58.772
		1.739	0.000	-5.595	0.000	0.000	0.000	65.136
		2.318	0.000	3.810	0.000	0.000	0.000	65.715
		2.898	0.000	11.525	0.000	0.000	0.000	61.442
		3.477	0.000	17.307	0.000	0.000	0.000	52.825
		4.057	0.000	21.398	0.000	0.000	0.000	41.455
		4.636	0.000	23.799	0.000	0.000	0.000	28.266
		5.216	0.000	24.508	0.000	0.000	0.000	14.191
		5.795	-0.000	23.528	-0.000	-0.000	-0.000	0.163
	107:0.9 (D+ F	0.000	0.000	-42.480	0.000	0.000	0.000	8.186
		0.580	0.000	-32.048	0.000	0.000	0.000	29.418
		1.159	0.000	-22.190	0.000	0.000	0.000	44.944
		1.739	0.000	-12.908	0.000	0.000	0.000	55.082
		2.318	0.000	-4.201	0.000	0.000	0.000	60.150
		2.898	0.000	3.930	0.000	0.000	0.000	60.464
		3.477	0.000	11.405	0.000	0.000	0.000	55.753

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		4.057	0.000	18.304	0.000	0.000	0.000	47.004
		4.636	0.000	24.627	0.000	0.000	0.000	34.533
		5.216	0.000	30.376	0.000	0.000	0.000	18.658
		5.795	-0.000	35.549	-0.000	-0.000	-0.000	-0.304
	108:0.9 (D+ F2	0.000	0.000	-44.011	0.000	0.000	0.000	8.656
		0.580	0.000	-33.158	0.000	0.000	0.000	30.638
		1.159	0.000	-22.914	0.000	0.000	0.000	46.688
		1.739	0.000	-13.280	0.000	0.000	0.000	57.142
		2.318	0.000	-4.255	0.000	0.000	0.000	62.336
		2.898	0.000	4.160	0.000	0.000	0.000	62.606
		3.477	0.000	11.878	0.000	0.000	0.000	57.683
		4.057	0.000	18.987	0.000	0.000	0.000	48.593
		4.636	0.000	25.486	0.000	0.000	0.000	35.674
		5.216	0.000	31.375	0.000	0.000	0.000	19.261
		5.795	-0.000	36.655	-0.000	-0.000	-0.000	-0.309
	109:1.2DL+1.2	0.000	0.000	-173.084	0.000	0.000	0.000	55.677
		0.580	0.000	-125.590	0.000	0.000	0.000	140.513
		1.159	0.000	-81.924	0.000	0.000	0.000	199.734
		1.739	0.000	-42.085	0.000	0.000	0.000	235.455
		2.318	0.000	-6.073	0.000	0.000	0.000	249.786
		2.898	0.000	26.112	0.000	0.000	0.000	244.839
		3.477	0.000	53.923	0.000	0.000	0.000	220.578
		4.057	0.000	77.907	0.000	0.000	0.000	181.792
		4.636	0.000	98.063	0.000	0.000	0.000	130.593
		5.216	0.000	114.392	0.000	0.000	0.000	69.094
		5.795	-0.000	126.894	-0.000	-0.000	-0.000	-0.592
	110:1.2DL+1.6\	0.000	0.000	-107.110	0.000	0.000	0.000	24.134
		0.580	0.000	-80.024	0.000	0.000	0.000	77.407
		1.159	0.000	-54.621	0.000	0.000	0.000	115.922
		1.739	0.000	-30.900	0.000	0.000	0.000	140.609
		2.318	0.000	-8.863	0.000	0.000	0.000	152.396
		2.898	0.000	11.492	0.000	0.000	0.000	152.212
		3.477	0.000	29.923	0.000	0.000	0.000	139.542
		4.057	0.000	46.672	0.000	0.000	0.000	116.990
		4.636	0.000	61.737	0.000	0.000	0.000	85.485
		5.216	0.000	75.120	0.000	0.000	0.000	45.958
		5.795	-0.000	86.820	-0.000	-0.000	-0.000	-0.665
	111:1.2DL+1.6\	0.000	0.000	-109.083	0.000	0.000	0.000	39.822
		0.580	0.000	-78.139	0.000	0.000	0.000	92.949
		1.159	0.000	-49.906	0.000	0.000	0.000	129.452
		1.739	0.000	-24.383	0.000	0.000	0.000	150.828
		2.318	0.000	-1.571	0.000	0.000	0.000	158.572
		2.898	0.000	18.530	0.000	0.000	0.000	154.180
		3.477	0.000	35.534	0.000	0.000	0.000	137.843
		4.057	0.000	49.828	0.000	0.000	0.000	112.736
		4.636	0.000	61.410	0.000	0.000	0.000	80.355
		5.216	0.000	70.283	0.000	0.000	0.000	42.197
		5.795	-0.000	76.444	-0.000	-0.000	-0.000	-0.244

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
	112:1.2DL+1.0	0.000	0.000	-107.759	0.000	0.000	0.000	24.123
		0.580	0.000	-80.519	0.000	0.000	0.000	77.722
		1.159	0.000	-54.977	0.000	0.000	0.000	116.481
		1.739	0.000	-31.132	0.000	0.000	0.000	141.337
		2.318	0.000	-8.984	0.000	0.000	0.000	153.227
		2.898	0.000	11.466	0.000	0.000	0.000	153.088
		3.477	0.000	29.976	0.000	0.000	0.000	140.407
		4.057	0.000	46.788	0.000	0.000	0.000	117.804
		4.636	0.000	61.903	0.000	0.000	0.000	86.217
		5.216	0.000	75.320	0.000	0.000	0.000	46.583
		5.795	-0.000	87.040	-0.000	-0.000	-0.000	-0.161
	113:1.2DL+1.0	0.000	0.000	-109.289	0.000	0.000	0.000	24.094
		0.580	0.000	-81.629	0.000	0.000	0.000	78.442
		1.159	0.000	-55.700	0.000	0.000	0.000	117.724
		1.739	0.000	-31.503	0.000	0.000	0.000	142.895
		2.318	0.000	-9.037	0.000	0.000	0.000	154.912
		2.898	0.000	11.696	0.000	0.000	0.000	154.729
		3.477	0.000	30.450	0.000	0.000	0.000	141.834
		4.057	0.000	47.472	0.000	0.000	0.000	118.891
		4.636	0.000	62.762	0.000	0.000	0.000	86.855
		5.216	0.000	76.320	0.000	0.000	0.000	46.683
		5.795	-0.000	88.146	-0.000	-0.000	-0.000	-0.670
3	100:1.4 (D + F	0.000	0.000	-7.073	0.000	0.000	0.000	-0.968
		0.027	0.000	-6.357	0.000	0.000	0.000	-0.784
		0.055	0.000	-5.642	0.000	0.000	0.000	-0.620
		0.082	0.000	-4.930	0.000	0.000	0.000	-0.475
		0.110	0.000	-4.219	0.000	0.000	0.000	-0.348
		0.137	0.000	-3.511	0.000	0.000	0.000	-0.241
		0.165	0.000	-2.805	0.000	0.000	0.000	-0.155
		0.192	0.000	-2.101	0.000	0.000	0.000	-0.088
		0.220	0.000	-1.398	0.000	0.000	0.000	-0.040
		0.247	0.000	-0.698	0.000	0.000	0.000	-0.011
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	101:1.2(D + F	0.000	0.000	-5.145	0.000	0.000	0.000	-0.703
		0.027	0.000	-4.623	0.000	0.000	0.000	-0.570
		0.055	0.000	-4.102	0.000	0.000	0.000	-0.450
		0.082	0.000	-3.583	0.000	0.000	0.000	-0.345
		0.110	0.000	-3.066	0.000	0.000	0.000	-0.253
		0.137	0.000	-2.551	0.000	0.000	0.000	-0.175
		0.165	0.000	-2.037	0.000	0.000	0.000	-0.113
		0.192	0.000	-1.525	0.000	0.000	0.000	-0.064
		0.220	0.000	-1.015	0.000	0.000	0.000	-0.029
		0.247	0.000	-0.507	0.000	0.000	0.000	-0.008
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	102:1.2(D + F	0.000	0.000	-2.119	0.000	0.000	0.000	-0.282
		0.027	0.000	-1.889	0.000	0.000	0.000	-0.227
		0.055	0.000	-1.662	0.000	0.000	0.000	-0.178
		0.082	0.000	-1.440	0.000	0.000	0.000	-0.136

Beam Force Detail Cont...

Beam	L/C	d	Axial	Shear		Torsion	Bending	
			Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.110	0.000	-1.222	0.000	0.000	0.000	-0.099
		0.137	0.000	-1.008	0.000	0.000	0.000	-0.068
		0.165	0.000	-0.798	0.000	0.000	0.000	-0.044
		0.192	0.000	-0.592	0.000	0.000	0.000	-0.025
		0.220	0.000	-0.391	0.000	0.000	0.000	-0.011
		0.247	0.000	-0.193	0.000	0.000	0.000	-0.003
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	103:1.2(D + F	0.000	0.000	-5.143	0.000	0.000	0.000	-0.703
		0.027	0.000	-4.620	0.000	0.000	0.000	-0.570
		0.055	0.000	-4.100	0.000	0.000	0.000	-0.450
		0.082	0.000	-3.581	0.000	0.000	0.000	-0.345
		0.110	0.000	-3.064	0.000	0.000	0.000	-0.253
		0.137	0.000	-2.549	0.000	0.000	0.000	-0.175
		0.165	0.000	-2.036	0.000	0.000	0.000	-0.112
		0.192	0.000	-1.524	0.000	0.000	0.000	-0.064
		0.220	0.000	-1.015	0.000	0.000	0.000	-0.029
		0.247	0.000	-0.506	0.000	0.000	0.000	-0.008
		0.275	-0.000	-0.000	-0.000	-0.000	-0.000	0.000
	104:1.2(D + F	0.000	0.000	-5.401	0.000	0.000	0.000	-0.739
		0.027	0.000	-4.854	0.000	0.000	0.000	-0.599
		0.055	0.000	-4.308	0.000	0.000	0.000	-0.473
		0.082	0.000	-3.764	0.000	0.000	0.000	-0.362
		0.110	0.000	-3.221	0.000	0.000	0.000	-0.266
		0.137	0.000	-2.680	0.000	0.000	0.000	-0.184
		0.165	0.000	-2.141	0.000	0.000	0.000	-0.118
		0.192	0.000	-1.603	0.000	0.000	0.000	-0.067
		0.220	0.000	-1.067	0.000	0.000	0.000	-0.031
		0.247	0.000	-0.533	0.000	0.000	0.000	-0.008
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	105:0.9 D + 1.(	0.000	0.000	-1.904	0.000	0.000	0.000	-0.258
		0.027	0.000	-1.707	0.000	0.000	0.000	-0.209
		0.055	0.000	-1.512	0.000	0.000	0.000	-0.165
		0.082	0.000	-1.317	0.000	0.000	0.000	-0.126
		0.110	0.000	-1.125	0.000	0.000	0.000	-0.092
		0.137	0.000	-0.933	0.000	0.000	0.000	-0.064
		0.165	0.000	-0.744	0.000	0.000	0.000	-0.041
		0.192	0.000	-0.556	0.000	0.000	0.000	-0.023
		0.220	0.000	-0.369	0.000	0.000	0.000	-0.011
		0.247	0.000	-0.184	0.000	0.000	0.000	-0.003
		0.275	-0.000	-0.000	-0.000	-0.000	-0.000	0.000
	106:0.9 D + 1.(	0.000	0.000	1.119	0.000	0.000	0.000	0.163
		0.027	0.000	1.025	0.000	0.000	0.000	0.134
		0.055	0.000	0.926	0.000	0.000	0.000	0.107
		0.082	0.000	0.824	0.000	0.000	0.000	0.083
		0.110	0.000	0.718	0.000	0.000	0.000	0.061
		0.137	0.000	0.608	0.000	0.000	0.000	0.043
		0.165	0.000	0.494	0.000	0.000	0.000	0.028
		0.192	0.000	0.376	0.000	0.000	0.000	0.016

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.220	0.000	0.255	0.000	0.000	0.000	0.007
		0.247	0.000	0.129	0.000	0.000	0.000	0.002
		0.275	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	107:0.9 (D+ F	0.000	0.000	-2.233	0.000	0.000	0.000	-0.304
		0.027	0.000	-2.004	0.000	0.000	0.000	-0.246
		0.055	0.000	-1.776	0.000	0.000	0.000	-0.194
		0.082	0.000	-1.549	0.000	0.000	0.000	-0.149
		0.110	0.000	-1.324	0.000	0.000	0.000	-0.109
		0.137	0.000	-1.100	0.000	0.000	0.000	-0.075
		0.165	0.000	-0.877	0.000	0.000	0.000	-0.048
		0.192	0.000	-0.656	0.000	0.000	0.000	-0.027
		0.220	0.000	-0.436	0.000	0.000	0.000	-0.012
		0.247	0.000	-0.217	0.000	0.000	0.000	-0.003
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	108:0.9 (D+ F	0.000	0.000	-2.270	0.000	0.000	0.000	-0.309
		0.027	0.000	-2.037	0.000	0.000	0.000	-0.250
		0.055	0.000	-1.805	0.000	0.000	0.000	-0.197
		0.082	0.000	-1.574	0.000	0.000	0.000	-0.151
		0.110	0.000	-1.345	0.000	0.000	0.000	-0.111
		0.137	0.000	-1.118	0.000	0.000	0.000	-0.076
		0.165	0.000	-0.891	0.000	0.000	0.000	-0.049
		0.192	0.000	-0.666	0.000	0.000	0.000	-0.028
		0.220	0.000	-0.443	0.000	0.000	0.000	-0.013
		0.247	0.000	-0.221	0.000	0.000	0.000	-0.003
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	109:1.2DL+1.2	0.000	0.000	-4.452	0.000	0.000	0.000	-0.592
		0.027	0.000	-3.968	0.000	0.000	0.000	-0.477
		0.055	0.000	-3.492	0.000	0.000	0.000	-0.375
		0.082	0.000	-3.024	0.000	0.000	0.000	-0.285
		0.110	0.000	-2.566	0.000	0.000	0.000	-0.208
		0.137	0.000	-2.115	0.000	0.000	0.000	-0.143
		0.165	0.000	-1.675	0.000	0.000	0.000	-0.091
		0.192	0.000	-1.243	0.000	0.000	0.000	-0.052
		0.220	0.000	-0.820	0.000	0.000	0.000	-0.023
		0.247	0.000	-0.406	0.000	0.000	0.000	-0.006
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	110:1.2DL+1.6\	0.000	0.000	-4.902	0.000	0.000	0.000	-0.665
		0.027	0.000	-4.394	0.000	0.000	0.000	-0.538
		0.055	0.000	-3.891	0.000	0.000	0.000	-0.425
		0.082	0.000	-3.391	0.000	0.000	0.000	-0.324
		0.110	0.000	-2.895	0.000	0.000	0.000	-0.238
		0.137	0.000	-2.402	0.000	0.000	0.000	-0.164
		0.165	0.000	-1.914	0.000	0.000	0.000	-0.105
		0.192	0.000	-1.430	0.000	0.000	0.000	-0.060
		0.220	0.000	-0.950	0.000	0.000	0.000	-0.027
		0.247	0.000	-0.473	0.000	0.000	0.000	-0.007
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	111:1.2DL+1.6\	0.000	0.000	-1.876	0.000	0.000	0.000	-0.244

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d	Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.027	0.000	-1.660	0.000	0.000	0.000	-0.195
		0.055	0.000	-1.451	0.000	0.000	0.000	-0.153
		0.082	0.000	-1.248	0.000	0.000	0.000	-0.116
		0.110	0.000	-1.050	0.000	0.000	0.000	-0.084
		0.137	0.000	-0.859	0.000	0.000	0.000	-0.057
		0.165	0.000	-0.675	0.000	0.000	0.000	-0.036
		0.192	0.000	-0.497	0.000	0.000	0.000	-0.021
		0.220	0.000	-0.325	0.000	0.000	0.000	-0.009
		0.247	0.000	-0.160	0.000	0.000	0.000	-0.002
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	112:1.2DL+1.0	0.000	0.000	-2.155	0.000	0.000	0.000	-0.161
		0.027	0.000	-1.677	0.000	0.000	0.000	-0.109
		0.055	0.000	-1.257	0.000	0.000	0.000	-0.069
		0.082	0.000	-0.894	0.000	0.000	0.000	-0.040
		0.110	0.000	-0.589	0.000	0.000	0.000	-0.020
		0.137	0.000	-0.341	0.000	0.000	0.000	-0.007
		0.165	0.000	-0.158	0.000	0.000	0.000	-0.000
		0.192	0.000	-0.033	0.000	0.000	0.000	0.002
		0.220	0.000	0.036	0.000	0.000	0.000	0.002
		0.247	0.000	0.046	0.000	0.000	0.000	0.001
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	113:1.2DL+1.0	0.000	0.000	-4.942	0.000	0.000	0.000	-0.670
		0.027	0.000	-4.430	0.000	0.000	0.000	-0.542
		0.055	0.000	-3.922	0.000	0.000	0.000	-0.428
		0.082	0.000	-3.417	0.000	0.000	0.000	-0.327
		0.110	0.000	-2.917	0.000	0.000	0.000	-0.239
		0.137	0.000	-2.421	0.000	0.000	0.000	-0.165
		0.165	0.000	-1.929	0.000	0.000	0.000	-0.106
		0.192	0.000	-1.441	0.000	0.000	0.000	-0.060
		0.220	0.000	-0.957	0.000	0.000	0.000	-0.027
		0.247	0.000	-0.476	0.000	0.000	0.000	-0.007
		0.275	-0.000	0.000	-0.000	-0.000	-0.000	-0.000

ATTACHMENT 2 F1 STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

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*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries6             *
*           Version  20.07.11.70                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    SEP 19, 2020                   *
*           Time=    15:37:44                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
*****************************************************
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```
     1. STAAD SPACE
INPUT FILE: D:\MAADEN\Flotation\STAAD\F1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-SEPT-20
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. JOB NO 49WF
     7. ENGINEER NAME BJB
     8. CHECKER NAME SJM
     9. APPROVED NAME AG
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 1 1.375 0 0; 2 1.65 0 0; 3 7.445 0 0; 7 7.72 0 0
    15. MEMBER INCIDENCES
    16. 1 1 2; 2 2 3; 3 3 7
    17. DEFINE MATERIAL START
    18. ISOTROPIC CONCRETE
    19. E 2.17185E+007
    20. POISSON 0.17
    21. DENSITY 25
    22. ALPHA 1E-005
    23. DAMP 0.05
    24. TYPE CONCRETE
    25. STRENGTH FCU 27579
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY AMERICAN
    28. 1 TO 3 PRIS YD 0.4 ZD 1.5
    29. CONSTANTS
    30. MATERIAL CONCRETE ALL
    31. SUPPORTS
    32. 2 3 FIXED BUT MY MZ KFY 0.2
    33. LOAD 100 LOADTYPE NONE  TITLE 1.4 ( D + F1)
    34. MEMBER LOAD
    35. 1 TO 3 UNI GY -23.64
    36. 1 TRAP GY 66.16 65.41
    37. 2 TRAP GY 65.41 49.73
    38. 3 TRAP GY 49.73 48.99
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39. JOINT LOAD
40. 2 FY -21.14 MZ 13.62
41. 3 FY -29.12
42. LOAD 101 LOADTYPE NONE  TITLE 1.2( D + F1 ) + 1.6 WLX
43. MEMBER LOAD
44. 1 TO 3 UNI GY -23.64
45. 1 TRAP GY 56.77 56.13
46. 2 TRAP GY 56.13 42.67
47. 3 TRAP GY 42.67 42.03
48. JOINT LOAD
49. 2 FY 4.3 MZ 11.7
50. 3 FY -47.7
51. LOAD 102 LOADTYPE NONE  TITLE 1.2( D + F1 ) + 1.6 WLZ
52. MEMBER LOAD
53. 1 TO 3 UNI GY -23.64
54. 1 TRAP GY 63.8 63.8
55. 2 TRAP GY 63.8 32.1
56. 3 TRAP GY 32.1 30.59
57. JOINT LOAD
58. 2 FY -8.5 MZ -27
59. 3 FY -25.6
60. LOAD 103 LOADTYPE NONE  TITLE 1.2( D + F2 ) + SLX
61. MEMBER LOAD
62. 1 TO 3 UNI GY -23.64
63. 1 TRAP GY 56.92 56.27
64. 2 TRAP GY 56.26 42.66
65. 3 TRAP GY 42.66 42.02
66. JOINT LOAD
67. 2 FY -18.2 MZ -11.8
68. 3 FY -25
69. LOAD 104 LOADTYPE NONE  TITLE 1.2( D + F2 ) + SLZ
70. MEMBER LOAD
71. 1 TO 3 UNI GY -23.64
72. 1 TRAP GY 56.86 56.26
73. 2 TRAP GY 56.26 43.58
74. 3 TRAP GY 43.58 42.98
75. JOINT LOAD
76. 2 FY -21.8 MZ -11
77. 3 FY -24.36
78. LOAD 105 LOADTYPE NONE  TITLE 0.9 D + 1.6 WLX
79. MEMBER LOAD
80. 1 TO 3 UNI GY -23.64
81. 1 TRAP GY 43.19 42.63
82. 2 TRAP GY 42.63 30.84
83. 3 TRAP GY 30.84 30.29
84. JOINT LOAD
85. 2 FY 10.7 MZ 10.2
86. 3 FY -41.2
87. LOAD 106 LOADTYPE NONE  TITLE 0.9 D + 1.6 WLZ
88. MEMBER LOAD
89. 1 TO 3 UNI GY -23.64
90. 1 TRAP GY 51.71 50.29
91. 2 TRAP GY 50.29 20.28
92. 3 TRAP GY 20.28 18.86
93. JOINT LOAD
94. 2 FY -2.1 MZ 26
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 95. 3 FY -19.2
 96. LOAD 107 LOADTYPE NONE  TITLE 0.9 ( D+ F2 ) + SLX
 97. MEMBER LOAD
 98. 1 TO 3 UNI GY -23.64
 99. 1 TRAP GY 42.69 42.21
100. 2 TRAP GY 42.21 32
101. 3 TRAP GY 32 31.52
102. JOINT LOAD
103. 2 FY -13.7 MZ 8.9
104. 3 FY -18.7
105. LOAD 108 LOADTYPE NONE  TITLE 0.9 ( D+ F2 ) + SLZ
106. MEMBER LOAD
107. 1 TO 3 UNI GY -23.64
108. 1 TRAP GY 43.49 42.97
109. 2 TRAP GY 42.97 32.15
110. 3 TRAP GY 32.15 31.64
111. JOINT LOAD
112. 2 FY -17 MZ 9.4
113. 3 FY -18.1
114. LOAD 109 LOADTYPE NONE  TITLE 1.2DL+1.2F1+1.6LL
115. MEMBER LOAD
116. 1 TO 3 UNI GY -23.64
117. 1 TRAP GY 112.59 109.39
118. 2 TRAP GY 109.37 41.44
119. 3 TRAP GY 41.44 38.22
120. JOINT LOAD
121. 2 FY -97 MZ 59
122. 3 FY -111
123. LOAD 110 LOADTYPE NONE  TITLE 1.2DL+1.6WLX+0.5LL
124. MEMBER LOAD
125. 1 TO 3 UNI GY -23.64
126. 1 TRAP GY 73.46 72.04
127. 2 TRAP GY 72.04 42.17
128. 3 TRAP GY 42.17 40.76
129. JOINT LOAD
130. 2 FY -18 MZ 26
131. 3 FY -1674
132. LOAD 111 LOADTYPE NONE  TITLE 1.2DL+1.6WLZ+0.5LL
133. MEMBER LOAD
134. 1 TO 3 UNI GY -23.64
135. 1 TRAP GY 81.99 79.71
136. 2 TRAP GY 79.71 31.6
137. 3 TRAP GY 31.6 29.32
138. JOINT LOAD
139. 2 FY -30 MZ 42
140. 3 FY -52
141. LOAD 112 LOADTYPE NONE  TITLE 1.2DL+1.0SLX+0.5LL+0.5F1
142. MEMBER LOAD
143. 1 TO 3 UNI GY -23.64
144. 1 TRAP GY 73.75 72.32
145. 2 TRAP GY 72.32 42.19
146. 3 TRAP GY 42.19 20.76
147. JOINT LOAD
148. 2 FY -41 MZ 26
149. 3 FY -52
150. LOAD 113 LOADTYPE NONE  TITLE 1.2DL+1.0SLZ+0.5LL+0.5F1
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  151. MEMBER LOAD
  152. 1 TO 3 UNI GY -23.64
  153. 1 TRAP GY 74.54 73.08
  154. 2 TRAP GY 73.08 42.34
  155. 3 TRAP GY 42.34 40.88
  156. JOINT LOAD
  157. 2 FY -45 MZ 26
  158. 3 FY -51
  159. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**4**	**NUMBER OF MEMBERS**	**3**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
  ORIGINAL/FINAL BAND-WIDTH=     1/     1/      9 DOF
TOTAL      PRIMARY LOAD CASES =    14, TOTAL DEGREES OF FREEDOM =      18
TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
  SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/ 324550.0 MB
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**NOTE - NEAR INSTABILITY AT JOINT     7   DIRECTION = FY
 PROBABLE CAUSE  - NEAR SINGULAR K-MATRIX
 K-MATRIX DIAG=  8.3885137E+04 L-MATRIX DIAG=  2.2839595E-03 EQN NO      14

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.
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  160. PRINT MEMBER FORCES ALL
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MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	100	1	0.00	0.00	0.00	0.00	0.00	-0.00
		2	0.00	-11.59	0.00	0.00	0.00	1.60
	101	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-9.02	0.00	0.00	0.00	1.24
	102	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-11.04	0.00	0.00	0.00	1.52
	103	1	0.00	-0.00	0.00	0.00	0.00	0.00
		2	0.00	-9.06	0.00	0.00	0.00	1.25
	104	1	0.00	-0.00	0.00	0.00	0.00	-0.00
		2	0.00	-9.05	0.00	0.00	0.00	1.25
	105	1	0.00	0.00	0.00	0.00	0.00	-0.00
		2	0.00	-5.30	0.00	0.00	0.00	0.73
	106	1	0.00	-0.00	0.00	0.00	0.00	0.00
		2	0.00	-7.52	0.00	0.00	0.00	1.04
	107	1	0.00	0.00	0.00	0.00	0.00	-0.00
		2	0.00	-5.17	0.00	0.00	0.00	0.71
	108	1	0.00	-0.00	0.00	0.00	0.00	0.00
		2	0.00	-5.39	0.00	0.00	0.00	0.74
	109	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-24.02	0.00	0.00	0.00	3.32
	110	1	0.00	0.00	0.00	0.00	0.00	-0.00
		2	0.00	-13.51	0.00	0.00	0.00	1.87
	111	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-15.73	0.00	0.00	0.00	2.18
	112	1	0.00	-0.00	0.00	0.00	0.00	-0.00
		2	0.00	-13.58	0.00	0.00	0.00	1.88
	113	1	0.00	-0.00	0.00	0.00	0.00	-0.00
		2	0.00	-13.80	0.00	0.00	0.00	1.91
2	100	2	0.00	-103.64	0.00	0.00	0.00	12.02
		3	0.00	-92.98	0.00	0.00	0.00	0.97
	101	2	0.00	-79.21	0.00	0.00	0.00	10.46
		3	0.00	-70.07	0.00	0.00	0.00	0.70
	102	2	0.00	-90.62	0.00	0.00	0.00	-28.52
		3	0.00	-50.26	0.00	0.00	0.00	0.28
	103	2	0.00	-83.51	0.00	0.00	0.00	-13.05
		3	0.00	-66.12	0.00	0.00	0.00	0.70
	104	2	0.00	-84.26	0.00	0.00	0.00	-12.25
		3	0.00	-68.04	0.00	0.00	0.00	0.74
	105	2	0.00	-41.96	0.00	0.00	0.00	9.47
		3	0.00	-33.93	0.00	0.00	0.00	0.26
	106	2	0.00	-43.96	0.00	0.00	0.00	24.96
		3	0.00	-23.53	0.00	0.00	0.00	-0.16

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
	107	2	0.00	-42.48	0.00	0.00	0.00	8.19
		3	0.00	-35.55	0.00	0.00	0.00	0.30
	108	2	0.00	-44.01	0.00	0.00	0.00	8.66
		3	0.00	-36.66	0.00	0.00	0.00	0.31
	109	2	0.00	-173.08	0.00	0.00	0.00	55.68
		3	0.00	-126.89	0.00	0.00	0.00	0.59
	110	2	0.00	-107.11	0.00	0.00	0.00	24.13
		3	0.00	-86.82	0.00	0.00	0.00	0.67
	111	2	0.00	-109.08	0.00	0.00	0.00	39.82
		3	0.00	-76.44	0.00	0.00	0.00	0.24
	112	2	0.00	-107.76	0.00	0.00	0.00	24.12
		3	0.00	-87.04	0.00	0.00	0.00	0.16
	113	2	0.00	-109.29	0.00	0.00	0.00	24.09
		3	0.00	-88.15	0.00	0.00	0.00	0.67
3	100	3	0.00	-7.07	0.00	0.00	0.00	-0.97
		7	0.00	-0.00	0.00	0.00	0.00	0.00
	101	3	0.00	-5.15	0.00	0.00	0.00	-0.70
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	102	3	0.00	-2.12	0.00	0.00	0.00	-0.28
		7	0.00	-0.00	0.00	0.00	0.00	0.00
	103	3	0.00	-5.14	0.00	0.00	0.00	-0.70
		7	0.00	0.00	0.00	0.00	0.00	0.00
	104	3	0.00	-5.40	0.00	0.00	0.00	-0.74
		7	0.00	-0.00	0.00	0.00	0.00	0.00
	105	3	0.00	-1.90	0.00	0.00	0.00	-0.26
		7	0.00	0.00	0.00	0.00	0.00	-0.00
	106	3	0.00	1.12	0.00	0.00	0.00	0.16
		7	0.00	0.00	0.00	0.00	0.00	0.00
	107	3	0.00	-2.23	0.00	0.00	0.00	-0.30
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	108	3	0.00	-2.27	0.00	0.00	0.00	-0.31
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	109	3	0.00	-4.45	0.00	0.00	0.00	-0.59
		7	0.00	-0.00	0.00	0.00	0.00	0.00
	110	3	0.00	-4.90	0.00	0.00	0.00	-0.67
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	111	3	0.00	-1.88	0.00	0.00	0.00	-0.24
		7	0.00	-0.00	0.00	0.00	0.00	0.00
	112	3	0.00	-2.15	0.00	0.00	0.00	-0.16
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	113	3	0.00	-4.94	0.00	0.00	0.00	-0.67
		7	0.00	-0.00	0.00	0.00	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

161. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 19,2020 TIME= 15:37:46 ****

****
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